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Exponential Approximation Space Based Finite Volume WENO Scheme for Solving the Five-equation Model of Compressible Two-medium Flows

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Abstract
In this work, we develop a finite volume weighted essentially non-oscillatory (WENO) scheme based on an exponential approximation space for solving the five-equation transport model of compressible two-medium flows. Compared to the scheme based on the conventional polynomial approximation space, the performance of the present scheme is improved by locally optimizing the shape parameter in the exponential function such that the reconstructed solution can attain one enhanced order of accuracy without the need of increasing the stencil size. More importantly, through analyzing the relationship between the shape parameter and the linear weights of the WENO approximation, we present a suitable upper bound for the shape parameter which is able to avoid the appearance of negative linear weights. Finally, with the aim of maintaining computational robustness under large density and pressure ratios, we design a bound-preserving and positivity-preserving limiting algorithm targeted for the non-polynomial approximation space based finite volume WENO scheme. Numerical results demonstrate that the present scheme achieves high-order accuracy, high resolution and strong robustness in simulating compressible two-medium flows.

Keywords Finite volume WENO scheme · Exponential approximation space · Five-equation transport model · Bound-preserving and positivity-preserving

Mathematics Subject Classification 35L40 · 65M99 · 76T10

1 Introduction

Compressible two-medium flows arise in a wide range of scientific and engineering fields, which generally possess the features of strong nonlinearity, large interface deformations and high density and pressure ratios. In recent years, the development of high-order numerical

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schemes for simulating compressible two-medium flows has gained many progresses, e.g. the finite volume weighted essentially non-oscillatory (WENO) scheme [1–4], finite difference WENO scheme [5–7] and discontinuous Galerkin (DG) method [8–13]. High accuracy is achieved by the schemes based on the conventional piecewise polynomial approximation space. However, since the polynomial functions are generally difficult to be regulated according to the local flow features, it may bring excessive numerical dissipations across discontinuities and regions with large gradients.

In order to achieve enhanced resolution in rapid transition regions, a promising approach is to develop numerical schemes based on non-polynomial approximation space. For example, Yuan and Shu [14] developed DG method based on non-polynomial approximation spaces consisting of exponential and trigonometric functions with the objective of obtaining better approximations for specific types of PDEs. Bigoni and Hesthaven [15] developed an adaptive finite volume WENO scheme based on the radial basis functions (RBFs) for hyperbolic conservation laws, where a constant shape parameter is introduced to control the balance between the accuracy and stability. Guo and Jung [16] proposed an adaptive finite difference WENO scheme based on the RBFs with a free parameter to improve the accuracy order. In [17–20], Yoon et al. proposed a finite difference WENO scheme based on the exponential approximation space. A shape parameter was used and determined by considering its relationship with the accuracy of the exponential polynomial interpolation, which results in an improved order of accuracy than the conventional polynomial interpolation without increasing the stencil size. Recently, Han et al. [21, 22] proposed a resolution-enhanced seventh-order finite difference WENO scheme which utilizes non-polynomial reconstructions by incorporating kriging interpolation and Gaussian exponential function. All of the aforementioned work provided promising numerical results in solving single-medium flow problems with more accurate reconstructions in the smooth regions and sharper solution profiles near the jump discontinuity. However, to the best of the authors' knowledge, their feasibility on the simulations of compressible two-medium flows is still unexplored.

Another critical challenge in the high-order simulations of compressible two-medium flows is the emergence of inadmissible numerical solutions, e.g. negative density, negative internal energy, and so forth. In fact, the emergence of inadmissible numerical solutions not only severely undermines the computational reliability, sometimes even leads to a breakdown of the computation. In order to preserve the intrinsic boundedness, Zhang and Shu generalized the maximum-principle-preserving and positivity-preserving techniques to construct high-order numerical methods for hyperbolic conservation laws [23–25] and compressible Euler equations [26, 27]. Its main idea is to exactly express the average value of the polynomial solution in each cell by utilizing the Gauss-Lobatto quadrature rule, and then preserve the intrinsic boundedness for the polynomial solution by repeated convex combinations. Inspired by this, Zhang and Cheng proposed a general framework for developing the bound-preserving and positivity-preserving high-order numerical schemes for solving the five-equation model [28] of compressible two-medium flows in a series of work [4, 10–13], which is well capable of maintaining the boundedness of volume fraction and the positivity of partial density, internal energy and squared speed of sound. Unfortunately, in the case of non-polynomial interpolation, the exactness of the Gauss-Lobatto quadrature rule is no longer satisfied. Therefore, specific analysis and strategy should be carried out in designing the bound-preserving and positivity-preserving limiting algorithm for the high-order numerical schemes based on non-polynomial approximation space.

In this work, we develop a finite volume WENO scheme based on an exponential approximation space for solving the five-equation transport model of compressible two-medium flows. The main contribution of this work reflects in the following three aspects: First, we

conduct a detailed study on the formulation of the shape parameter used in the exponential function such that it leads to an enhanced order of accuracy at each quadrature point targeted for the finite volume WENO framework; Second, based on the relationship between the shape parameter and the linear weights of the finite volume WENO approximation, we present a theoretical analysis of the upper bound for the shape parameter in order to maintain non-negative linear weights during the computation; Third, we propose a specific strategy for keeping the consistency between the cell-averaged solution and the convex combination of the non-polynomial reconstructed solutions at the quadrature points, which is essential in designing the bound-preserving and positivity-preserving limiting algorithm for non-polynomial approximation space based numerical schemes. The high-order accuracy, high resolution and strong robustness properties of the present scheme are well demonstrated by a series of benchmark numerical tests. The numerical results validate that compared to the finite volume WENO scheme based on the conventional polynomial approximation space, the present scheme exhibits better performance in simulating compressible two-medium flows, especially in the regions near the material interface.

This paper is organized as follows. The finite volume WENO scheme for the five-equation transport model is described in Section 2. The exponential approximation space based WENO reconstruction associated with the specific formulation of the shape parameter is investigated in Section 3. Section 4 presents the bound-preserving and positivity-preserving limiting algorithms in both one and two dimensions. Numerical results for compressible gas-gas and gas-water two-medium flows are presented in Section 5. Finally, we provide the concluding remarks in Section 6.

2 Finite Volume WENO Scheme for the Five-equation Transport Model

This work considers the five-equation transport model of compressible two-medium flows [28]

$$\frac{\partial \mathbf{Q}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{Q}) = (\nabla \cdot \mathbf{u})\mathbf{z}, \tag{1}$$

where

$$\mathbf{Q} = \begin{bmatrix} z_1\rho_1 \\ z_2\rho_2 \\ \rho\mathbf{u} \\ E \\ z_1 \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = [\mathbf{F}_x(\mathbf{Q}), \mathbf{F}_y(\mathbf{Q})] = \begin{bmatrix} z_1\rho_1\mathbf{u}^{\mathrm{T}} \\ z_2\rho_2\mathbf{u}^{\mathrm{T}} \\ \rho\mathbf{u}\mathbf{u}^{\mathrm{T}} + p\mathbb{I}_d \\ (E+p)\mathbf{u}^{\mathrm{T}} \\ z_1\mathbf{u}^{\mathrm{T}} \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ z_1 \end{bmatrix}. \tag{2}$$

Here, the solution domain is $\Omega \in \mathbb{R}^d$ with $d = 1, 2$ being the spatial dimension, $z_k\rho_k > 0$ and $z_k \in [0, 1]$ are the partial density and volume fraction of the kth $(k = 1, 2)$ medium, respectively, $\mathbf{u}$ is the velocity vector, where $\mathbf{u} = u$ and $\mathbf{u} = (u, v)^{\mathrm{T}}$ for one and two dimensions, respectively, $E = \rho e + \rho|\mathbf{u}|^2/2$ is the total energy, ρ, p and e are the mixture density, pressure and specific internal energy, respectively, which are determined by the following mixture rules

$$\rho = z_1\rho_1 + z_2\rho_2, \quad p = z_1p_1 + z_2p_2, \quad \rho e = z_1\rho_1e_1 + z_2\rho_2e_2, \quad 1 = z_1 + z_2, \tag{3}$$

where ρ_k, p_k and e_k are the phase density, pressure and specific internal energy of the kth medium, respectively. Their relationship is given by the stiffened gas equation of state (EoS)

$$\rho_k e_k = \frac{p_k + \gamma_k \pi_{\infty,k}}{\gamma_k - 1}, \tag{4}$$

where $\gamma_k > 1$ and $\pi_{\infty,k}$ are both material depended constants. If $\pi_{\infty,k} = 0$, then the stiffened gas EoS (4) reduces to the ideal gas EoS. Based on the mechanical equilibrium $p = p_1 = p_2$, multiplying (4) by z_k and then summing over both mediums, we obtain the mixture EoS

$$\rho e = \frac{p + \gamma \pi_\infty}{\gamma - 1}, \tag{5}$$

where γ and π_∞ are properties of the mixture which are defined by the following relations

$$\frac{1}{\gamma - 1} = \frac{z_1}{\gamma_1 - 1} + \frac{z_2}{\gamma_2 - 1}, \quad \frac{\gamma \pi_\infty}{\gamma - 1} = \frac{z_1 \gamma_1 \pi_{\infty,1}}{\gamma_1 - 1} + \frac{z_2 \gamma_2 \pi_{\infty,2}}{\gamma_2 - 1}. \tag{6}$$

The mixture speed of sound, which is denoted as c, is expressed as follows

$$c^2 = \frac{\gamma(p + \pi_\infty)}{\rho} = \frac{\gamma(\gamma - 1)(\rho e - \pi_\infty)}{\rho}. \tag{7}$$

We assume that Ω_h is an approximation of Ω, which contains a set of non-overlapping uniform rectangular cells $\Omega_h = \{I_{i,j} | 1 \le i \le N_x, 1 \le j \le N_y\}$ with $I_{i,j} = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}]$. For a given cell $I_{i,j} \in \Omega_h$, the cell center is $(x_i, y_j) = \big((x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}})/2, (y_{j-\frac{1}{2}} + y_{j+\frac{1}{2}})/2\big)$, cell sizes are $\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$ and $\Delta y = y_{j+\frac{1}{2}} - y_{j-\frac{1}{2}}$, and cell area is $|I_{i,j}| = \Delta x \Delta y$. We denote the approximate solution on $I_{i,j}$ as $\mathbf{Q}_{i,j}(x, y)$ with the cell average defined as

$$\overline{\mathbf{Q}}_{i,j} = \frac{1}{|I_{i,j}|} \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{Q}_{i,j}(x, y) \mathrm{d}x \mathrm{d}y. \tag{8}$$

Besides, we denote $\mathbf{Q}_{i\pm\frac{1}{2},j}(y)$ and $\mathbf{Q}_{i,j\pm\frac{1}{2}}(x)$ as the traces of the approximate solution $\mathbf{Q}_{i,j}(x, y)$ on the four cell edges $x = x_{i\pm\frac{1}{2}}$ and $y = y_{j\pm\frac{1}{2}}$, respectively.

Integrating the five-equation transport model (1) over the cell $I_{i,j} \in \Omega_h$, it gives the equation

$$\frac{\mathrm{d}\overline{\mathbf{Q}}_{i,j}}{\mathrm{d}t} + \frac{1}{|I_{i,j}|} \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \nabla \cdot \mathbf{F}(\mathbf{Q}_{i,j}) \mathrm{d}x \mathrm{d}y = \frac{1}{|I_{i,j}|} \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} (\nabla \cdot \mathbf{u}_{i,j}) \mathbf{z}_{i,j} \mathrm{d}x \mathrm{d}y. \tag{9}$$

Using the cell-averaged solution $\overline{\mathbf{z}}_{i,j}$ to approximate $\mathbf{z}_{i,j}$ and then performing integration by parts for the conservative term, we obtain the semi-discrete formulation

$$\begin{aligned} &\frac{\mathrm{d}\overline{\mathbf{Q}}_{i,j}}{\mathrm{d}t} + \frac{1}{\Delta x} \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} \widehat{\mathbf{F}}_x(\mathbf{Q}^-_{i+\frac{1}{2},j}, \mathbf{Q}^+_{i+\frac{1}{2},j}) - \widehat{\mathbf{F}}_x(\mathbf{Q}^-_{i-\frac{1}{2},j}, \mathbf{Q}^+_{i-\frac{1}{2},j}) \mathrm{d}y \\ &\quad + \frac{1}{\Delta y} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \widehat{\mathbf{F}}_y(\mathbf{Q}^-_{i,j+\frac{1}{2}}, \mathbf{Q}^+_{i,j+\frac{1}{2}}) - \widehat{\mathbf{F}}_y(\mathbf{Q}^-_{i,j-\frac{1}{2}}, \mathbf{Q}^+_{i,j-\frac{1}{2}}) \mathrm{d}x \\ &\quad = \Big(\frac{1}{\Delta x} \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} (\widehat{u}_{i+\frac{1}{2},j} - \widehat{u}_{i-\frac{1}{2},j}) \mathrm{d}y + \frac{1}{\Delta y} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} (\widehat{v}_{i,j+\frac{1}{2}} - \widehat{v}_{i,j-\frac{1}{2}}) \mathrm{d}x\Big) \overline{\mathbf{z}}_{i,j}, \end{aligned} \tag{10}$$

where $\mathbf{Q}^{\pm}_{i\pm\frac{1}{2},j}(y)$ and $\mathbf{Q}^{\pm}_{i,j\pm\frac{1}{2}}(x)$ are the WENO reconstructed solutions along the cell interfaces $x = x_{i\pm\frac{1}{2}}$ and $y = y_{j\pm\frac{1}{2}}$, respectively, $\widehat{\mathbf{F}}_x$ and $\widehat{\mathbf{F}}_y$ are the numerical flux of the conservative term, and $\widehat{u}_{i\pm\frac{1}{2},j}$ and $\widehat{v}_{i,j\pm\frac{1}{2}}$ are the numerical flux of the source term.

The numerical flux of the conservative term is computed by the Lax-Friedrichs flux [29]

$$\begin{aligned}\widehat{\mathbf{F}}_x(\mathbf{Q}_L,\mathbf{Q}_R) &= \frac{1}{2}\big(\mathbf{F}_x(\mathbf{Q}_L)+\mathbf{F}_x(\mathbf{Q}_R)-S_x(\mathbf{Q}_R-\mathbf{Q}_L)\big),\\ \widehat{\mathbf{F}}_y(\mathbf{Q}_L,\mathbf{Q}_R) &= \frac{1}{2}\big(\mathbf{F}_y(\mathbf{Q}_L)+\mathbf{F}_y(\mathbf{Q}_R)-S_y(\mathbf{Q}_R-\mathbf{Q}_L)\big),\end{aligned} \tag{11}$$

where S_x and S_y are the estimated maximum characteristic speed along x- and y-directions, respectively, and their definitions will be specified later. The numerical flux of the source term is closely related to that of the conservative term. Specifically, the numerical flux of the source term, which is consistent with that of the conservative term (11), is defined as follows

$$\widehat{u}(u_L,u_R) = \frac{1}{2}(u_L+u_R), \quad \widehat{v}(v_L,v_R) = \frac{1}{2}(v_L+v_R). \tag{12}$$

The integrals in (10) are calculated by the four-point Gauss-Lobatto quadrature rule, which exactly integrates the polynomials of degree equal to or less than five. Specifically, the four-point Gauss-Lobatto quadrature points and quadrature weights in the interval $[-1,1]$ are given by

$$\widehat{x} \in \{-1, -\frac{\sqrt{5}}{5}, \frac{\sqrt{5}}{5}, 1\}, \quad \widehat{\omega} \in \{\frac{1}{6}, \frac{5}{6}, \frac{5}{6}, \frac{1}{6}\}. \tag{13}$$

The quadrature rule at two dimensions is given by a tensor product of the above one-dimensional quadrature rule, and in the following we denote the quadrature points for the cell $I_{i,j} \in \Omega_h$ as $(\widehat{x}_i^{\alpha}, \widehat{y}_j^{\beta})$ $(\alpha,\beta = 1,\cdots,4)$. This completes the spatial discretization for the model equations.

The temporal evolution of the numerical scheme is conducted by the third-order strong stability preserving Runge-Kutta (SSP-RK) scheme [30]

$$\begin{aligned}\overline{\mathbf{Q}}^{n_1}_{i,j} &= \overline{\mathbf{Q}}^{n}_{i,j} + \Delta t\mathbf{R}(\overline{\mathbf{Q}}^{n}_{i,j}),\\ \overline{\mathbf{Q}}^{n_2}_{i,j} &= \frac{3}{4}\overline{\mathbf{Q}}^{n}_{i,j} + \frac{1}{4}\overline{\mathbf{Q}}^{n_1}_{i,j} + \frac{1}{4}\Delta t\mathbf{R}(\overline{\mathbf{Q}}^{n_1}_{i,j}),\\ \overline{\mathbf{Q}}^{n+1}_{i,j} &= \frac{1}{3}\overline{\mathbf{Q}}^{n}_{i,j} + \frac{2}{3}\overline{\mathbf{Q}}^{n_2}_{i,j} + \frac{2}{3}\Delta t\mathbf{R}(\overline{\mathbf{Q}}^{n_2}_{i,j}),\end{aligned} \tag{14}$$

where the superscript 'n' denotes the time instant $t = t^n$ $(n \in \mathbb{N})$, $\Delta t = t^{n+1} - t^n$ is the time step and $\mathbf{R}(\cdot)$ denotes the residual vector of the spatial discretization. Then, it is easy to prove that the finite volume WENO scheme (10) associated with third-order SSP-RK scheme (14) satisfies two important criterions of compressible two-medium flow simulations, that is

- (Single-medium consistency) If $\overline{(z_1)}^{n}_{i,j} \equiv 1,\ \forall I_{i,j} \in \Omega_h$, then $\overline{(z_1)}^{n+1}_{i,j} \equiv 1,\ \forall I_{i,j} \in \Omega_h$;
- (Mechanical equilibrium [31]) If $\overline{\mathbf{u}}^{n}_{i,j} \equiv \mathbf{u}_0$ and $\overline{p}^{n}_{i,j} \equiv p_0,\ \forall I_{i,j} \in \Omega_h$, then $\overline{\mathbf{u}}^{n+1}_{i,j} \equiv \mathbf{u}_0$ and $\overline{p}^{n+1}_{i,j} \equiv p_0,\ \forall I_{i,j} \in \Omega_h$.

It should be mentioned that the well preservation of the mechanical equilibrium around an isolated material interface is important for maintaining an oscillation-free material interface.

3 Exponential Approximation Space Based WENO Reconstruction

3.1 Exponential Approximation Space

In the following, we present the details of the one-dimensional exponential approximation space based WENO reconstruction. Consider the target cell $I_j = [x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}] \in \Omega_h$ and the global stencil for the WENO reconstruction $S_5 = \{I_{j+l} | l = -2, \cdots, 2\}$, the associated exponential approximation space is expressed as follows

$$\mathbb{E}_5(x) = \text{span}\{1,\ x - x_j,\ (x - x_j)^2,\ e^{\lambda(x-x_j)},\ e^{-\lambda(x-x_j)}\} \doteq \text{span}\{\phi_m^j(x) | m = 0, \cdots, 4\}, \tag{15}$$

where λ is the shape parameter which will be specified later. When $\lambda \neq 0$, the Wronskian matrix of the functions $\phi_m^j(x) \in \mathbb{E}_5(x)$ is non-singular which means they are linear independent. We denote each component of $\mathbf{Q}_j(x)$ as $q_j(x)$, and define the reconstructed solution in $\mathbb{E}_5(x)$ as

$$q_j(x) = a_0 + a_1(x - x_j) + a_2(x - x_j)^2 + a_3 e^{\lambda(x-x_j)} + a_4 e^{-\lambda(x-x_j)}, \tag{16}$$

where the reconstructed coefficients a_m are determined through enforcing the following condition

$$\frac{1}{\Delta x} \int_{I_{j+l}} q_j(x) \mathrm{d}x = \overline{q}_{j+l}. \tag{17}$$

Taking $x = x_{i\pm\frac{1}{2}}$ into (16), it gives the reconstructed solutions at the two cell interfaces

$$\begin{aligned} q^+_{j-\frac{1}{2}} &= q_j(\widehat{x}_j^1) = a_{0,1}\overline{q}_{j-2} + a_{1,1}\overline{q}_{j-1} + a_{2,1}\overline{q}_j + a_{3,1}\overline{q}_{j+1} + a_{4,1}\overline{q}_{j+2}, \\ q^-_{j+\frac{1}{2}} &= q_j(\widehat{x}_j^4) = a_{0,4}\overline{q}_{j-2} + a_{1,4}\overline{q}_{j-1} + a_{2,4}\overline{q}_j + a_{3,4}\overline{q}_{j+1} + a_{4,4}\overline{q}_{j+2}, \end{aligned} \tag{18}$$

where the truncated forms of the coefficients $a_{m,1}$ and $a_{m,4}$ are expressed as follows

$$\begin{aligned} a_{0,1} = a_{4,4} &= -\frac{3}{60} + \frac{8(\lambda\Delta x)^2}{840} - \frac{6(\lambda\Delta x)^4}{5040} + \mathcal{O}(\Delta x^6), \\ a_{1,1} = a_{3,4} &= \frac{27}{60} - \frac{18(\lambda\Delta x)^2}{840} + \frac{13(\lambda\Delta x)^4}{5040} + \mathcal{O}(\Delta x^6), \\ a_{2,1} = a_{2,4} &= \frac{47}{60} + \frac{6(\lambda\Delta x)^2}{840} - \frac{3(\lambda\Delta x)^4}{5040} + \mathcal{O}(\Delta x^6), \\ a_{3,1} = a_{1,4} &= -\frac{13}{60} + \frac{10(\lambda\Delta x)^2}{840} - \frac{9(\lambda\Delta x)^4}{5040} + \mathcal{O}(\Delta x^6), \\ a_{4,1} = a_{0,4} &= \frac{2}{60} - \frac{6(\lambda\Delta x)^2}{840} + \frac{5(\lambda\Delta x)^4}{5040} + \mathcal{O}(\Delta x^6). \end{aligned} \tag{19}$$

Next, we divide the global stencil S_5 into three sub-stencils $S_5^{[\ell]} = \{I_{j+\ell-2}, I_{j+\ell-1}, I_{j+\ell}\}$ ($\ell = 0, 1, 2$), and reconstruct the approximate solution in each sub-stencil by using the conventional polynomial approximation space [32, 33], that is $q_j^{[\ell]}(x) \in \mathbb{P}_3(x)$ with

$$\mathbb{P}_3(x) = \text{span}\{1,\ (x - x_j),\ (x - x_j)^2\}. \tag{20}$$

Denoting the local reconstructed solution at $x = x_{j\pm\frac{1}{2}}$ as $q^{\mp,[\ell]}_{j\pm\frac{1}{2}}$, then the global reconstructed solution (18) can be decomposed into a linear combination of the local reconstructed solutions

$$\begin{aligned} q^{+}_{j-\frac{1}{2}} &= d_1^{[0]} q^{+,[0]}_{j-\frac{1}{2}} + d_1^{[1]} q^{+,[1]}_{j-\frac{1}{2}} + d_1^{[2]} q^{+,[2]}_{j-\frac{1}{2}}, \\ q^{-}_{j+\frac{1}{2}} &= d_4^{[0]} q^{-,[0]}_{j+\frac{1}{2}} + d_4^{[1]} q^{-,[1]}_{j+\frac{1}{2}} + d_4^{[2]} q^{-,[2]}_{j+\frac{1}{2}}, \end{aligned} \tag{21}$$

where $d_\alpha^{[\ell]}$ are referred to as the ℓth linear weight at $x = \widehat{x}_j^\alpha$ computed by

$$\begin{aligned} d_1^{[0]} = d_4^{[2]} &= \frac{3}{10} - \frac{8(\lambda\Delta x)^2}{140} + \frac{12(\lambda\Delta x)^4}{1680} + \mathcal{O}(\Delta x^6), \\ d_1^{[1]} = d_4^{[1]} &= \frac{6}{10} + \frac{11(\lambda\Delta x)^2}{140} - \frac{17(\lambda\Delta x)^4}{1680} + \mathcal{O}(\Delta x^6), \\ d_1^{[2]} = d_4^{[0]} &= \frac{1}{10} - \frac{3(\lambda\Delta x)^2}{140} + \frac{5(\lambda\Delta x)^4}{1680} + \mathcal{O}(\Delta x^6). \end{aligned} \tag{22}$$

As $\Delta x \to 0$, the linear weights $(d_1^{[0]}, d_1^{[1]}, d_1^{[2]})$ based on the exponential polynomials converge to the optimal linear weights $(\frac{3}{10}, \frac{6}{10}, \frac{1}{10})$ based on the conventional polynomials [32, 33].

Following a similar way, we can also derive the relationships between the global and local reconstructed solutions at the two inner quadrature points $x = x_{j\pm\frac{\sqrt{5}}{10}}$, that is

$$\begin{aligned} q_{j-\frac{\sqrt{5}}{10}} &= q_j(\widehat{x}_j^2) = d_2^{[0]} q^{[0]}_{j-\frac{\sqrt{5}}{10}} + d_2^{[1]} q^{[1]}_{j-\frac{\sqrt{5}}{10}} + d_2^{[2]} q^{[2]}_{j-\frac{\sqrt{5}}{10}}, \\ q_{j+\frac{\sqrt{5}}{10}} &= q_j(\widehat{x}_j^3) = d_3^{[0]} q^{[0]}_{j+\frac{\sqrt{5}}{10}} + d_3^{[1]} q^{[1]}_{j+\frac{\sqrt{5}}{10}} + d_3^{[2]} q^{[2]}_{j+\frac{\sqrt{5}}{10}}, \end{aligned} \tag{23}$$

where the linear weights are given by

$$\begin{aligned} d_2^{[0]} = d_3^{[2]} &= \frac{91 - 9\sqrt{5}}{440} - \frac{20257 - 1789\sqrt{5}}{462000}(\lambda\Delta x)^2 \\ &+ \frac{3671283 - 292765\sqrt{5}}{609840000}(\lambda\Delta x)^4 + \mathcal{O}(\Delta x^6), \\ d_2^{[1]} = d_3^{[1]} &= \frac{258}{440} + \frac{20257}{231000}(\lambda\Delta x)^2 - \frac{15893}{1320000}(\lambda\Delta x)^4 + \mathcal{O}(\Delta x^6), \\ d_2^{[2]} = d_3^{[0]} &= \frac{91 + 9\sqrt{5}}{440} - \frac{20257 + 1789\sqrt{5}}{462000}(\lambda\Delta x)^2 \\ &+ \frac{3671283 + 292765\sqrt{5}}{609840000}(\lambda\Delta x)^4 + \mathcal{O}(\Delta x^6). \end{aligned} \tag{24}$$

Similar to the WENO reconstruction based on the polynomial approximation space, in order to avoid spurious oscillations around the discontinuities, the nonlinear weights which are determined by measuring the smoothness of the local solution on each sub-stencil are introduced. Here, we apply a modified WENO-Z type nonlinear weights [34, 35] which are computed as follows

$$\varpi_\alpha^{[\ell]} := \frac{\tau_\alpha^{[\ell]}}{\sum_{\ell=0}^{2} \tau_\alpha^{[\ell]}}, \quad \tau_\alpha^{[\ell]} := d_\alpha^{[\ell]}\left(1 + \frac{\tau_5^2}{\beta^{[\ell]^2} + \varepsilon}\right). \tag{25}$$

We refer to [18] for more details on the formulations of τ_5 and $\beta^{[\ell]}$ which are specially designed to ensure the accuracy $|\varpi_\alpha^{[\ell]} - d_\alpha^{[\ell]}| = \mathcal{O}(\Delta x^4)$.

Remark 1 The extension of the exponential approximation space based WENO reconstruction to two dimensions can be implemented through using a dimension-by-dimension strategy. Specifically, for the target cell $I_{i,j} \in \Omega_h$ and the global reconstructed stencil $S_5 = \{I_{i+l,j+r} | l, r = -2, \cdots, 2\}$, we first derive the global reconstructed solutions $q_{i+l,j}(y) \in \mathbb{E}_5(y)$ and local reconstructed solutions $q^{[\ell]}_{i+l,j}(y) \in \mathbb{P}_3(y)$ along y-direction through enforcing

$$\frac{1}{\Delta y}\int_{y_{j+r-\frac{1}{2}}}^{y_{j+r+\frac{1}{2}}} q_{i+l,j}(y)\mathrm{d}y = \overline{q}_{i+l,j+r}, \quad \frac{1}{\Delta y}\int_{y_{j+s-\frac{1}{2}}}^{y_{j+s+\frac{1}{2}}} q^{[\ell]}_{i+l,j}(y)\mathrm{d}y = \overline{q}_{i+l,j+s}, \tag{26}$$

where $s = \ell - 2, \ell - 1, \ell$, and then compute the reconstructed solutions at the quadrature points

$$q_{i+l,j}(\widehat{y}^{\beta}_j) = \varpi^{[0]}_{\beta} q^{[0]}_{i+l,j}(\widehat{y}^{\beta}_j) + \varpi^{[1]}_{\beta} q^{[1]}_{i+l,j}(\widehat{y}^{\beta}_j) + \varpi^{[2]}_{\beta} q^{[2]}_{i+l,j}(\widehat{y}^{\beta}_j). \tag{27}$$

Next, we derive the global reconstructed solution $q_{i,j}(x, \widehat{y}^{\beta}_j) \in \mathbb{E}_5(x)$ and local reconstructed solutions $q^{[\ell]}_{i,j}(x, \widehat{y}^{\beta}_j) \in \mathbb{P}_3(x)$ along x-direction through enforcing

$$\frac{1}{\Delta x}\int_{x_{i+l-\frac{1}{2}}}^{x_{i+l+\frac{1}{2}}} q_{i,j}(x, \widehat{y}^{\beta}_j)\mathrm{d}x = q_{i+l,j}(\widehat{y}^{\beta}_j), \quad \frac{1}{\Delta x}\int_{x_{i+s-\frac{1}{2}}}^{x_{i+s+\frac{1}{2}}} q^{[\ell]}_{i,j}(x, \widehat{y}^{\beta}_j)\mathrm{d}x = q_{i+s,j}(\widehat{y}^{\beta}_j). \tag{28}$$

Finally, the reconstructed solution at each two-dimensional quadrature point is computed by

$$q_{i,j}(\widehat{x}^{\alpha}_i, \widehat{y}^{\beta}_j) = \varpi^{[0]}_{\alpha} q^{[0]}_{i,j}(\widehat{x}^{\alpha}_i, \widehat{y}^{\beta}_j) + \varpi^{[1]}_{\alpha} q^{[1]}_{i,j}(\widehat{x}^{\alpha}_i, \widehat{y}^{\beta}_j) + \varpi^{[2]}_{\alpha} q^{[2]}_{i,j}(\widehat{x}^{\alpha}_i, \widehat{y}^{\beta}_j). \tag{29}$$

Remark 2 In order to eliminate the spurious oscillations around an isolated material interface, the exponential approximation space based WENO reconstruction is performed in characteristic space by applying a cell-wise characteristic projection [11]. Moreover, it can be proved that the mechanical equilibrium is well maintained through using the cell-wise characteristic projection. We refer to Appendix A for the detailed procedure of implementation.

3.2 Shape Parameter Formulation

This subsection introduces the formulation of the shape parameter λ. Assuming $q(x) \in \mathbf{C}^5(\Omega)$ that are smooth enough, then taking the Taylor expansion of $\overline{q}_{j+l}$ around each quadrature point and substituting it into the reconstructed solutions (21) and (23), we have

$$\begin{aligned} q^{+}_{j-\frac{1}{2}} &= q(x_{j-\frac{1}{2}}) + \frac{1}{60}\big(q^{(5)}(x_{j-\frac{1}{2}}) - \lambda^2 q^{(3)}(x_{j-\frac{1}{2}})\big)\Delta x^5 + \mathcal{O}(\Delta x^6), \\ q^{-}_{j+\frac{1}{2}} &= q(x_{j+\frac{1}{2}}) - \frac{1}{60}\big(q^{(5)}(x_{j+\frac{1}{2}}) - \lambda^2 q^{(3)}(x_{j+\frac{1}{2}})\big)\Delta x^5 + \mathcal{O}(\Delta x^6), \end{aligned} \tag{30}$$

and

$$\begin{aligned} q_{j-\frac{\sqrt{5}}{10}} &= q(x_{j-\frac{\sqrt{5}}{10}}) + \frac{383\sqrt{5}}{90000}\big(q^{(5)}(x_{j-\frac{\sqrt{5}}{10}}) - \lambda^2 q^{(3)}(x_{j-\frac{\sqrt{5}}{10}})\big)\Delta x^5 + \mathcal{O}(\Delta x^6), \\ q_{j+\frac{\sqrt{5}}{10}} &= q(x_{j+\frac{\sqrt{5}}{10}}) - \frac{383\sqrt{5}}{90000}\big(q^{(5)}(x_{j+\frac{\sqrt{5}}{10}}) - \lambda^2 q^{(3)}(x_{j+\frac{\sqrt{5}}{10}})\big)\Delta x^5 + \mathcal{O}(\Delta x^6). \end{aligned} \tag{31}$$

Here, the subscript '(υ)' denotes the υth derivative of the function. The above expressions imply that with a suitable choice of λ, the accuracy of the reconstructed solutions for smooth problems can enhance one additional order. That is the following Lemma [17, 18].

Table 1 Numerical interpolation and difference of $q^{(r)}(\widehat{x}_j^\alpha)$

$\widehat{x}_j^\alpha$	$q^{(r)}(\widehat{x}_j^\alpha)$
$x_{j-\frac{1}{2}}$	$q^{(3)} \approx \frac{1}{6\Delta x^3}(\overline{q}_{j-3} - 11\overline{q}_{j-2} + 28\overline{q}_{j-1} - 28\overline{q}_j + 11\overline{q}_{j+1} - \overline{q}_{j+2})$
	$q^{(5)} \approx \frac{1}{\Delta x^5}(-\overline{q}_{j-3} + 5\overline{q}_{j-2} - 10\overline{q}_{j-1} + 10\overline{q}_j - 5\overline{q}_{j+1} + \overline{q}_{j+2})$
$x_{j-\frac{\sqrt{5}}{10}}$	$q^{(3)} \approx \frac{1}{30\Delta x^3}\big[8\overline{q}_{j-3} - (55+3\sqrt{5})\overline{q}_{j-2} + (110+12\sqrt{5})\overline{q}_{j-1}$
	$-(80+18\sqrt{5})\overline{q}_j + (10+12\sqrt{5})\overline{q}_{j+1} + (7-3\sqrt{5})\overline{q}_{j+2}\big]$
	$q^{(5)} \approx \frac{1}{\Delta x^5}(-\overline{q}_{j-3} + 5\overline{q}_{j-2} - 10\overline{q}_{j-1} + 10\overline{q}_j - 5\overline{q}_{j+1} + \overline{q}_{j+2})$
$x_{j+\frac{\sqrt{5}}{10}}$	$q^{(3)} \approx \frac{1}{30\Delta x^3}\big[-(7-3\sqrt{5})\overline{q}_{j-2} - (10+12\sqrt{5})\overline{q}_{j-1} + (80+18\sqrt{5})\overline{q}_j$
	$-(110+12\sqrt{5})\overline{q}_{j+1} + (55+3\sqrt{5})\overline{q}_{j+2} - 8\overline{q}_{j+3}\big]$
	$q^{(5)} \approx \frac{1}{\Delta x^5}(-\overline{q}_{j-2} + 5\overline{q}_{j-1} - 10\overline{q}_j + 10\overline{q}_{j+1} - 5\overline{q}_{j+2} + \overline{q}_{j+3})$
$x_{j+\frac{1}{2}}$	$q^{(3)} \approx \frac{1}{6\Delta x^3}(\overline{q}_{j-2} - 11\overline{q}_{j-1} + 28\overline{q}_j - 28\overline{q}_{j+1} + 11\overline{q}_{j+2} - \overline{q}_{j+3})$
	$q^{(5)} \approx \frac{1}{\Delta x^5}(-\overline{q}_{j-2} + 5\overline{q}_{j-1} - 10\overline{q}_j + 10\overline{q}_{j+1} - 5\overline{q}_{j+2} + \overline{q}_{j+3})$

Lemma 1 *Consider the problem with smooth solution $q(x) \in \mathbf{C}^5(\Omega)$ and $|q^{(3)}(x)| \geq \varepsilon$ for some $\varepsilon > 0$, if the shape parameter λ of the exponential approximation space $\mathbb{E}_5(x)$ at the Gauss-Lobatto quadrature point $\widehat{x}_j^\alpha$ $(\alpha = 1, \cdots, 4)$ of any cell $I_j \in \Omega_h$ is chosen as*

$$\lambda_\alpha^2 = \frac{q^{(5)}(\widehat{x}_j^\alpha)}{q^{(3)}(\widehat{x}_j^\alpha)} + \mathcal{O}(\Delta x), \tag{32}$$

then the exponential approximation space based WENO reconstructions (30) and (31) can achieve the optimal sixth-order of accuracy.

In practical computation, the high-order derivatives in (32) are calculated by the numerical interpolation and difference. Specifically, we first choose a six-cell stencil around each quadrature point and reconstruct a fifth-order polynomial solution $\widetilde{q}(x) = q(x) + \mathcal{O}(\Delta x^6)$ through preserving the average value of each cell. Then, as listed in Tab. 1, the derivatives $q^{(3)}(x)$ and $q^{(5)}(x)$ at the quadrature points are approximated by $\widetilde{q}^{(3)}(x)$ and $\widetilde{q}^{(5)}(x)$, respectively, and the shape parameter that satisfies the accuracy requirement in Lemma 1 is calculated as follows

$$\lambda_\alpha^2 = \frac{\widetilde{q}^{(5)}(\widehat{x}_j^\alpha)}{\widetilde{q}^{(3)}(\widehat{x}_j^\alpha)} = \frac{q^{(5)}(\widehat{x}_j^\alpha) + \mathcal{O}(\Delta x)}{q^{(3)}(\widehat{x}_j^\alpha) + \mathcal{O}(\Delta x^3)} = \frac{q^{(5)}(\widehat{x}_j^\alpha)}{q^{(3)}(\widehat{x}_j^\alpha)} + \mathcal{O}(\Delta x). \tag{33}$$

For compressible two-medium flows involving large density and pressure ratios, the solutions around an isolated material interface may contain large gradients. Under this circumstance, the absolute value of the shape parameter computed by (32) or (33) may become very large to reflect the sharp variation of the solutions. However, an extremely large shape parameter may cause the linear weights to be negative and leads to a nonlinear instability. Thus, a suitable upper bound for the shape parameter should be set in the computation. According to the expressions of the linear weights at $x = x_{j\pm\frac{1}{2}}$ (22), through neglecting the higher-order infinitesimal terms and maintaining the remaining terms to be non-negative, we

derive the following three inequalities

$$\begin{cases} \frac{12}{1680}(\lambda\Delta x)^4 - \frac{8}{140}(\lambda\Delta x)^2 + \frac{3}{10} \geq 0, \\ \frac{17}{1680}(\lambda\Delta x)^4 - \frac{11}{140}(\lambda\Delta x)^2 - \frac{6}{10} \leq 0, \\ \frac{5}{1680}(\lambda\Delta x)^4 - \frac{3}{140}(\lambda\Delta x)^2 + \frac{1}{10} \geq 0. \end{cases} \tag{34}$$

The restriction on λ at $x = x_{j\pm\frac{1}{2}}$ is determined by solving the above inequalities. Note that the expressions on the left side of (34) can be viewed as a quadratic function of $(\lambda\Delta x)^2$. Moreover, it can be verified that the discriminants of first and third inequalities are negative, which implies that the two inequalities always hold true. Therefore, the value interval of λ at $x = x_{j\pm\frac{1}{2}}$ is directly determined by solving the second inequality, the solution of which is given by

$$0 < |\lambda_{j\pm\frac{1}{2}}\Delta x|^2 \leq |\frac{66 - 6\sqrt{597}}{17}| \simeq 4.74. \tag{35}$$

Similarly, by enforcing the linear wights at $x = x_{j\pm\frac{\sqrt{5}}{10}}$ (24) satisfy the following three inequalities

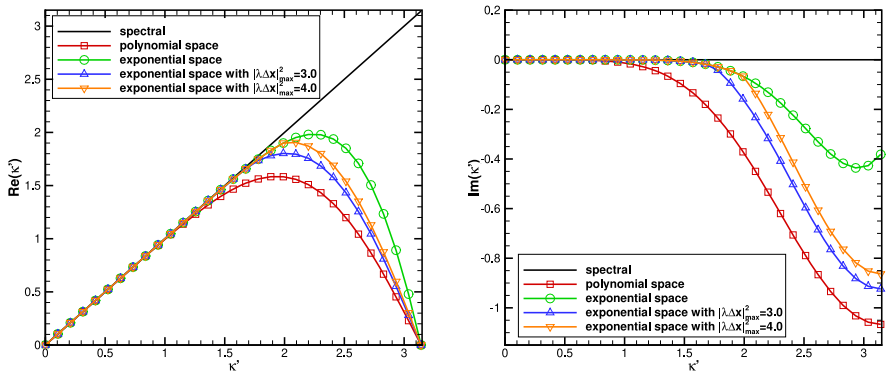
$$\begin{cases} \frac{3671283 - 292765\sqrt{5}}{609840000}(\lambda\Delta x)^4 - \frac{20257 - 1789\sqrt{5}}{462000}(\lambda\Delta x)^2 + \frac{91 - 9\sqrt{5}}{440} \geq 0, \\ \frac{15893}{1320000}(\lambda\Delta x)^4 - \frac{20257}{231000}(\lambda\Delta x)^2 - \frac{258}{440} \leq 0, \\ \frac{3671283 + 292765\sqrt{5}}{609840000}(\lambda\Delta x)^4 - \frac{20257 + 1789\sqrt{5}}{462000}(\lambda\Delta x)^2 + \frac{91 + 9\sqrt{5}}{440} \geq 0, \end{cases} \tag{36}$$

we derive the restriction on λ at $x = x_{j\pm\frac{\sqrt{5}}{10}}$ as follows

$$0 < |\lambda_{j\pm\frac{\sqrt{5}}{10}}\Delta x|^2 \leq 4.23. \tag{37}$$

Combining (35) and (37), we prudently choose the upper bound for the shape parameters taken at the four Gauss-Lobatto quadrature points as $|\lambda\Delta x|^2_{\max} \in [3.0, 4.0]$ in the computation.

Finally, in order to study the influence of the upper bound for the shape parameter $|\lambda\Delta x|^2_{\max}$ on the numerical scheme, we perform a Fourier analysis to study the spectral behavior of the present scheme in solving the one-dimensional scalar linear advection equation. Here, the detailed analysis procedure is inspired by the way presented in [36, 37]. Fig. 1 shows the dispersion and dissipation properties of the present exponential approximation space based finite volume WENO schemes with and without using $|\lambda\Delta x|^2_{\max}$. The spectral behavior of the finite volume WENO scheme based on the conventional polynomial approximation space is also shown in this figure for the purpose of comparison. We can observe the exponential approximation space based scheme without using $|\lambda\Delta x|^2_{\max}$ exhibits better performance than the one using $|\lambda\Delta x|^2_{\max}$. Specifically, the dispersion and dissipation errors increase with the decrease of the upper bound value. However, for both using and without using $|\lambda\Delta x|^2_{\max}$, the exponential approximation space based schemes produce lower dispersion and dissipation errors than the polynomial approximation space based scheme. This clearly demonstrates that the effectiveness of the exponential approximation space in enhancing the computational resolution.



(a) dispersion (b) dissipation

Fig. 1 Spectral property of the finite volume WENO scheme

4 Bound-Preserving and Positivity-Preserving Limiting Algorithm

The five-equation transport model is hyperbolic provided that the mixture speed of sound (7) is well defined [28]. Thus, the set of admissible states for the model equations is given by

$$\mathcal{Q} = \{\mathbf{Q} = [z_1\rho_1, z_2\rho_2, \rho\mathbf{u}^{\mathrm{T}}, E, z_1]^{\mathrm{T}} | z_1\rho_1 > 0, z_2\rho_2 > 0, \rho c^2 > 0, 0 \le z_1 \le 1\}. \quad (38)$$

According to (7), the above set can be rewritten into the following equivalent form

$$\mathcal{Q} = \{\mathbf{Q} = [z_1\rho_1, z_2\rho_2, \rho\mathbf{u}^{\mathrm{T}}, E, z_1]^{\mathrm{T}} | z_1\rho_1 > 0, z_2\rho_2 > 0, \rho e - \pi_\infty > 0, 0 \le z_1 \le 1\}. \quad (39)$$

Then, we have the following lemma [4].

Lemma 2 *If $\gamma_2 \ge \gamma_1 \ge 1$ ($\gamma_1 \ge \gamma_2 \ge 1$) and $\pi_{\infty,2} \ge \pi_{\infty,1} \ge 0$ ($\pi_{\infty,1} \ge \pi_{\infty,2} \ge 0$), then the admissible set $\mathcal{Q}$ is a convex set, i.e. for any $\mathbf{Q}_1$, $\mathbf{Q}_2 \in \mathcal{Q}$ and $\theta \in [0, 1]$, it gives*

$$\theta\mathbf{Q}_1 + (1-\theta)\mathbf{Q}_2 \in \mathcal{Q}. \quad (40)$$

Inspired by the work of Zhang and Shu [23, 24, 26], the authors established a general framework of designing the bound-preserving and positivity-preserving limiting algorithm for compressible two-medium flows in previous studies [4, 10–13]. The main idea is to keep the updated cell average $\overline{\mathbf{Q}}_j^{n+1} \in \mathcal{Q}$ by repeated convex combinations with the use of the Gauss-Lobatto quadrature rule for calculating $\overline{\mathbf{Q}}_j^n$ associated with a scaling limiter for modifying the high-order reconstructed solution $\mathbf{Q}_h^n(x)$. However, in the case of non-polynomial interpolation, e.g. the exponential polynomial interpolation considered in this work, a potential issue lies in that the cell average $\overline{\mathbf{Q}}_j^n$ cannot be expressed exactly by using the conventional Gauss-Lobatto quadrature rule.

According to the expression of the reconstructed solution $q_j(x)$ (16), in order to guarantee that the cell average $\overline{q}_j$ can still be decomposed into a convex combination of the reconstructed solutions at the quadrature points, the following equality is required to be satisfied

$$\overline{q}_j = \sum_{\alpha=1}^{4} \frac{\widehat{\omega}_\alpha}{2} q_j(\widehat{x}_j^\alpha) = \sum_{m=0}^{4} a_m \Big(\sum_{\alpha=1}^{4} \frac{\widehat{\omega}_\alpha}{2} \phi_m^j(\widehat{x}_j^\alpha) \Big). \quad (41)$$

With the aim of preserving the above equality, instead of calculating the integrals of the exponential polynomials $\phi_m^j(x) \in \mathbb{E}_5(x)$ in (17) exactly, the four-point Gauss-Lobatto quadrature rule is applied in the integral calculations. Then, the reconstructed coefficients are determined by solving the following system of linear equations

$$\begin{bmatrix} \Phi_{j-2}^0 & \Phi_{j-2}^1 & \Phi_{j-2}^2 & \Phi_{j-2}^3 & \Phi_{j-2}^4 \\ \Phi_{j-1}^0 & \Phi_{j-1}^1 & \Phi_{j-1}^2 & \Phi_{j-1}^3 & \Phi_{j-1}^4 \\ \Phi_{j}^0 & \Phi_{j}^1 & \Phi_{j}^2 & \Phi_{j}^3 & \Phi_{j}^4 \\ \Phi_{j+1}^0 & \Phi_{j+1}^1 & \Phi_{j+1}^2 & \Phi_{j+1}^3 & \Phi_{j+1}^4 \\ \Phi_{j+2}^0 & \Phi_{j+2}^1 & \Phi_{j+2}^2 & \Phi_{j+2}^3 & \Phi_{j+2}^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} \overline{q}_{j-2} \\ \overline{q}_{j-1} \\ \overline{q}_{j} \\ \overline{q}_{j+1} \\ \overline{q}_{j+2} \end{bmatrix}, \tag{42}$$

where

$$\Phi_{j+l}^m = \sum_{\alpha=1}^{4} \frac{\widehat{\omega}_\alpha}{2} \phi_m^j(\widehat{x}_{j+l}^\alpha), \quad \forall I_{j+l} \in S_5. \tag{43}$$

Note that the use of the four-point Gauss-Lobatto quadrature rule guarantees that

$$\Phi_{j+l}^m = \frac{1}{\Delta x} \int_{I_{j+l}} \phi_m^j(x) \mathrm{d}x + \mathcal{O}(\Delta x^6), \tag{44}$$

and thus the accuracy order of the reconstructed solution calculated by (42) and(43) is still well maintained. More importantly, through this way we can keep the convex decomposition of the cell average as given by (41) hold true. This establishes the essential foundation for extending the bound-preserving and positivity-preserving limiting algorithm targeted for polynomial approximation space based schemes to those based on non-polynomial approximation space.

Since only two reconstructed solutions $\mathbf{Q}_{j\pm\frac{1}{2}}^{\mp}$ are explicitly used in the one-dimensional finite volume WENO scheme, we further rewrite the cell average into the following form

$$\overline{\mathbf{Q}}_j = \frac{\widehat{\omega}_1}{2} \mathbf{Q}_{j-\frac{1}{2}}^{+} + (1 - \frac{\widehat{\omega}_1}{2} - \frac{\widehat{\omega}_4}{2}) \mathbf{Q}_j(\widehat{x}_j^*) + \frac{\widehat{\omega}_4}{2} \mathbf{Q}_{j+\frac{1}{2}}^{-}, \tag{45}$$

where

$$\mathbf{Q}_j(\widehat{x}_j^*) = \frac{\overline{\mathbf{Q}}_j - \frac{\widehat{\omega}_1}{2} \mathbf{Q}_{j-\frac{1}{2}}^{+} - \frac{\widehat{\omega}_4}{2} \mathbf{Q}_{j+\frac{1}{2}}^{-}}{1 - \frac{\widehat{\omega}_1}{2} - \frac{\widehat{\omega}_4}{2}}. \tag{46}$$

From the intermediate value theorem, we have that $\widehat{x}_j^* \in I_j$. Then, similar to the finite volume WENO scheme based on the conventional polynomial approximation space [4], we can derive the sufficient condition for maintaining one-dimensional bounded and positive cell averages computed by the finite volume WENO scheme based on the exponential approximation space.

Theorem 1 *Consider the exponential approximation space based finite volume WENO scheme combined with the third-order strong stability preserving Runge-Kutta scheme for solving the one-dimensional five-equation transport model, if* $\overline{\mathbf{Q}}_j^n \in \mathcal{Q}$ *and* $\mathbf{Q}_j^n(\widehat{x}_j^\alpha) \in \mathcal{Q}$, $\forall \widehat{x}_j^\alpha \in \{\widehat{x}_j^1, \widehat{x}_j^*, \widehat{x}_j^4\}$ *and* $\forall I_j \in \Omega_h$, *then* $\overline{\mathbf{Q}}_j^{n+1} \in \mathcal{Q}$ *under the following time step restriction*

$$S \frac{\Delta t}{\Delta x} \leq \frac{\widehat{\omega}_1}{4} \min \left\{ 1, \frac{\rho e(\overline{\mathbf{Q}}_j^n) - \pi_\infty(\overline{\mathbf{Q}}_j^n)}{\pi_\infty(\overline{\mathbf{Q}}_j^n)} \right\}, \tag{47}$$

where S is the estimated maximum characteristic speed defined as

$$S \geq |u^{n,\mp}_{j\pm\frac{1}{2}}| + \max\{c(\mathbf{Q}^{n,\mp}_{j\pm\frac{1}{2}}), \tilde{c}(\mathbf{Q}^{n,\mp}_{j\pm\frac{1}{2}})\} \ \ with \ \ \tilde{c}(\mathbf{Q}) = \sqrt{\frac{p^2}{2\rho(\rho e - \pi_\infty)}}. \tag{48}$$

Next, we extend the above theorem to two dimensions. By using the integral calculations (42) and (43), the two-dimensional cell average of the cell $I_{i,j} \in \Omega_h$ can be expressed as

$$\overline{\mathbf{Q}}_{i,j} = \sum_{\alpha=1}^{4}\sum_{\beta=1}^{4} \frac{\widehat{\omega}_\alpha \widehat{\omega}_\beta}{4} \mathbf{Q}_{i,j}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta) = \sum_{\partial I_{i,j}} \frac{\widehat{\omega}_\alpha \widehat{\omega}_\beta}{4} \mathbf{Q}_{i,j}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta) + (1 - \sum_{\partial I_{i,j}} \frac{\widehat{\omega}_\alpha \widehat{\omega}_\beta}{4}) \mathbf{Q}_{i,j}(\widehat{x}_i^*, \widehat{y}_j^*), \tag{49}$$

where $\partial I_{i,j}$ denotes the union of the cell interfaces of $I_{i,j}$, and $(\widehat{x}_i^*, \widehat{y}_j^*) \in I_{i,j}$ that satisfies

$$\mathbf{Q}_{i,j}(\widehat{x}_i^*, \widehat{y}_j^*) = \frac{\overline{\mathbf{Q}}_{i,j} - \sum_{\partial I_{i,j}} \frac{\widehat{\omega}_\alpha \widehat{\omega}_\beta}{4} \mathbf{Q}_{i,j}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta)}{1 - \sum_{\partial I_{i,j}} \frac{\widehat{\omega}_\alpha \widehat{\omega}_\beta}{4}}. \tag{50}$$

Then, the sufficient condition for maintaining two-dimensional bounded and positive cell averages of the present scheme is given by the following theorem.

Theorem 2 *Consider the exponential approximation space based finite volume WENO scheme combined with the third-order strong stability preserving Runge-Kutta scheme for solving the two-dimensional five-equation transport model, if $\overline{\mathbf{Q}}^n_{i,j} \in \mathcal{Q}$ and $\mathbf{Q}^n_{i,j}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta) \in \mathcal{Q}$, $\forall (\widehat{x}_i^\alpha, \widehat{y}_j^\beta) \in \partial I_{i,j} \cup (\widehat{x}_i^*, \widehat{y}_j^*)$ and $\forall I_{i,j} \in \Omega_h$, then $\overline{\mathbf{Q}}^{n+1}_{i,j} \in \mathcal{Q}$ under the following time step restriction*

$$S\Big(\frac{\Delta t}{\Delta x} + \frac{\Delta t}{\Delta y}\Big) \leq \frac{\widehat{\omega}_1}{4} \min\{1, \frac{\rho e(\overline{\mathbf{Q}}^n_{i,j}) - \pi_\infty(\overline{\mathbf{Q}}^n_{i,j})}{\pi_\infty(\overline{\mathbf{Q}}^n_{i,j})}\}, \quad \forall I_{i,j} \in \Omega_h, \tag{51}$$

where $S = \max_{\alpha,\beta}\{S_x^\beta, S_y^\alpha\}$ is the estimated maximum characteristic speed with S_x^β and S_y^α being

$$S_x^\beta \geq |u^{n,\mp}_{i\pm\frac{1}{2},\beta}| + \max\{c(\mathbf{Q}^{n,\mp}_{i\pm\frac{1}{2},\beta}), \tilde{c}(\mathbf{Q}^{n,\mp}_{i\pm\frac{1}{2},\beta})\}, \ \ S_y^\alpha \geq |v^{n,\mp}_{\alpha,j\pm\frac{1}{2}}| + \max\{c(\mathbf{Q}^{n,\mp}_{\alpha,j\pm\frac{1}{2}}), \tilde{c}(\mathbf{Q}^{n,\mp}_{\alpha,j\pm\frac{1}{2}})\}, \tag{52}$$

and $\mathbf{Q}^{n,\mp}_{i\pm\frac{1}{2},\beta} = \mathbf{Q}^n_{i,j}(x_{i\pm\frac{1}{2}}, \widehat{y}_j^\beta)$ and $\mathbf{Q}^{n,\mp}_{\alpha,j\pm\frac{1}{2}} = \mathbf{Q}^n_{i,j}(\widehat{x}_i^\alpha, y_{j\pm\frac{1}{2}})$.

The proof of Theorem 1 and Theorem 2 can be conducted following a similar way as those presented in the authors' previous work [4], and thus is omitted here. As stated by the two theorems, in order to maintain bounded and positive cell averages, the reconstructed solutions at the quadrature points should be inside the admissible set $\mathcal{Q}$ (39). This is achieved by applying a bound-preserving and positivity-preserving limiter to each targeted reconstructed solution. Here, we present the specific procedure at two dimensions in Algorithm 1, and the implementation of the one-dimensional case is similar. Finally, we mention that the convex decomposition of the cell average as given by (41) and (49) are still valid with the use of the cell-wise characteristic projection procedure, and we refer to Appendix A for a detailed proving process.

Algorithm 1 Two-dimensional bound-preserving and positivity-preserving limiter.

Require: $\overline{\mathbf{Q}}^n_{i,j} \in \mathcal{Q},\ \forall I_{i,j} \in \Omega_h$

Ensure: $\mathbf{Q}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) \in \mathcal{Q},\ \forall(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) \in \partial I_{i,j} \cup (\widehat{x}^*_i, \widehat{y}^*_j)$

1: Compute the reconstructed solutions $\mathbf{Q}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j)$ at $(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) \in \partial I_{i,j} \cup (\widehat{x}^*_i, \widehat{y}^*_j)$;

2: **if** $\mathbf{Q}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) \notin \mathcal{Q}$ **then**

3: if $(z_k\rho_k)^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) < \varepsilon$, then compute the scaling factor for $z_k\rho_k$ by

$$\theta^\rho_{\alpha,\beta} = \min\{1, \frac{\overline{(z_k\rho_k)}^n_{i,j} - \varepsilon}{\overline{(z_k\rho_k)}^n_{i,j} - (z_k\rho_k)^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j)}\};$$

4: if $(z_1)^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) \notin [\varepsilon, 1-\varepsilon]$, then compute the scaling factor for z_1 by

$$\theta^z_{\alpha,\beta} = \begin{cases} \min\{1, \dfrac{\overline{(z_1)}^n_{i,j} - \varepsilon}{\overline{(z_1)}^n_{i,j} - (z_1)^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j)}\}, & \text{if } (z_1)^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) < \varepsilon, \\ \min\{1, \dfrac{\overline{(z_1)}^n_{i,j} - (1-\varepsilon)}{\overline{(z_1)}^n_{i,j} - (z_1)^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j)}\}, & \text{if } (z_1)^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) > 1-\varepsilon; \end{cases}$$

5: if $c^2(\mathbf{Q}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j)) < \varepsilon$, then compute the scaling factor for c^2 by

$$\theta^c_{\alpha,\beta} = \min\{1, \frac{\rho e(\overline{\mathbf{Q}}^n_{i,j}) - \pi_\infty(\overline{\mathbf{Q}}^n_{i,j}) - \varepsilon}{\rho e(\overline{\mathbf{Q}}^n_{i,j}) - \pi_\infty(\overline{\mathbf{Q}}^n_{i,j}) - \big(\rho e(\mathbf{Q}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j)) - \pi_\infty(\mathbf{Q}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j))\big)}\};$$

6: **end if**

7: Set the final scaling factor by $\theta_{\alpha,\beta} = \min\{\theta^\rho_{\alpha,\beta}, \theta^z_{\alpha,\beta}, \theta^c_{\alpha,\beta}\}$;

8: Modify the reconstructed solutions into

$$\widetilde{\mathbf{Q}}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j) = (1-\theta_{\alpha,\beta})\overline{\mathbf{Q}}^n_{i,j} + \theta_{\alpha,\beta}\mathbf{Q}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j).$$

9: **return** $\widetilde{\mathbf{Q}}^n_{i,j}(\widehat{x}^\alpha_i, \widehat{y}^\beta_j)$

5 Numerical Tests

In this section, several benchmark and challenging one- and two-dimensional test cases will be investigated in order to demonstrate the effectiveness of the present scheme in the simulations of compressible two-medium flows. In the following, we simply denote the present finite volume WENO scheme based on the exponential approximation space as 'WENO-EXP' scheme. For comparison purpose, we also conduct the computation by using the finite volume WENO schemes based on the conventional polynomial approximation space, including the WENO-JS [32, 33] and WENO-Z [34, 35] schemes. Unless otherwise stated, the bound-preserving and positivity-preserving limiter and the cell-wise characteristic projection are used in the computation.

5.1 One-dimensional Smooth Advection Problem

The first case is a one-dimensional smooth advection problem with the initial condition given by

$$(\rho_1, \rho_2, u, p, z_1) = (1.0, 2.0, 1.0, 1.0, 0.5 + 0.4999\sin(2\pi x)), \tag{53}$$

where $x \in [0, 1]$ with periodic boundary conditions, and the output time is $T = 1.0$. The material parameters are taken as $\gamma_1 = 1.4,\ \pi_{\infty,1} = 0.0$ and $\gamma_2 = 1.6,\ \pi_{\infty,2} = 7.5$.

We conduct the computation with four different mesh resolutions $N = 20, 40, 80, 160$ to investigate the numerical errors of the WENO-JS, WENO-Z and WENO-EXP schemes. The results listed in Tab. 2 demonstrate that the WENO-EXP scheme produces lower numerical errors than the other two schemes, and it is able to achieve the designed sixth-order of accuracy in solving the one-dimensional smooth advection problem.

Next, we conduct the computation again by the WENO-EXP scheme without using the cell-wise characteristic projection (CCP). Fig. 2 displays the numerical results of the WENO-EXP schemes with and without using CCP. We can see that although the density profiles obtained by the two schemes agree well with each other, the errors of velocity and pressure profiles are quite different. Specifically, the velocity and pressure errors of the WENO-EXP scheme with CCP are significantly lower than those of the scheme without CCP. This demonstrates the mechanical equilibrium preserving property of the present CCP based WENO-EXP scheme, which is essential for avoiding the spurious oscillations around an isolated material interface.

5.2 One-dimensional Sharp Advection Problem

In this case, we consider the advection of an isolated sharp material interface in one dimension. The initial conditions are specified as follows

$$(\rho_1, \rho_2, u, p, z_1) = \begin{cases} (1.0, 0.1, 1.0, 1.0, 10^{-8}), & \text{if} \quad 0.4 \le x \le 0.6, \\ (1.0, 0.1, 1.0, 1.0, 1.0 - 10^{-8}), & \text{otherwise}, \end{cases} \tag{54}$$

where $x \in [0, 1]$ with periodic boundary conditions, and the output time is $T = 1.0$. The material parameters are taken as $\gamma_1 = 1.67,\ \pi_{\infty,1} = 7.5$ and $\gamma_2 = 1.4,\ \pi_{\infty,2} = 0.0$. The solution domain is uniformly divided into $N = 500$ mesh cells.

First, we present the comparison of the numerical results, which are computed by the WENO-JS, WENO-Z and WENO-EXP schemes, with the exact solution in Fig. 3. It shows that compared to the WENO-JS and WENO-Z schemes, the present WENO-EXP scheme provides an obviously sharper isolated material interface due to the use of the exponential approximation space, which possesses the capability of resolving the numerical solutions with large gradients.

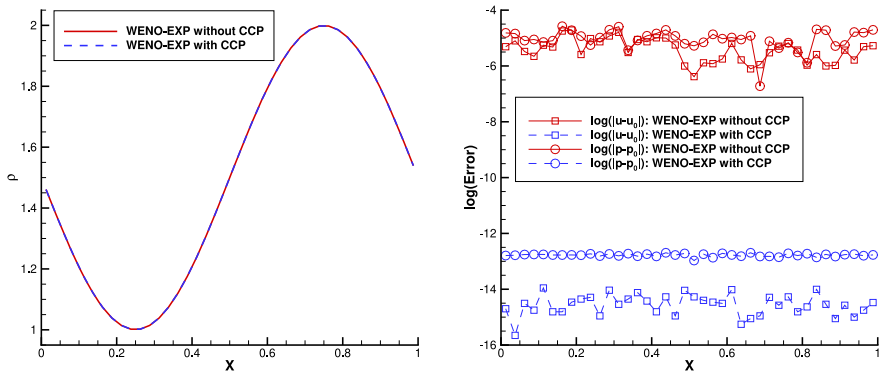
Next, in order to test the performance of the bound-preserving and positivity-preserving (BPPP) limiter, we conduct the computation again by using the WENO-EXP scheme without BPPP limiter. Fig. 4 shows the bound errors for $z_k\rho_k$ and z_1, which are defined as

$$\mathrm{BE}(z_k\rho_k) = \begin{cases} -z_k\rho_k, & \text{if} \quad z_k\rho_k < 0, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathrm{BE}(z_1) = \begin{cases} z_1 - 1, & \text{if} \quad z_1 > 1, \\ -z_1, & \text{if} \quad z_1 < 0, \\ 0, & \text{otherwise}. \end{cases} \tag{55}$$

We can see that the WENO-EXP scheme without BPPP limiter generates inadmissible numerical solutions around the material interface, including negative partial densities and volume

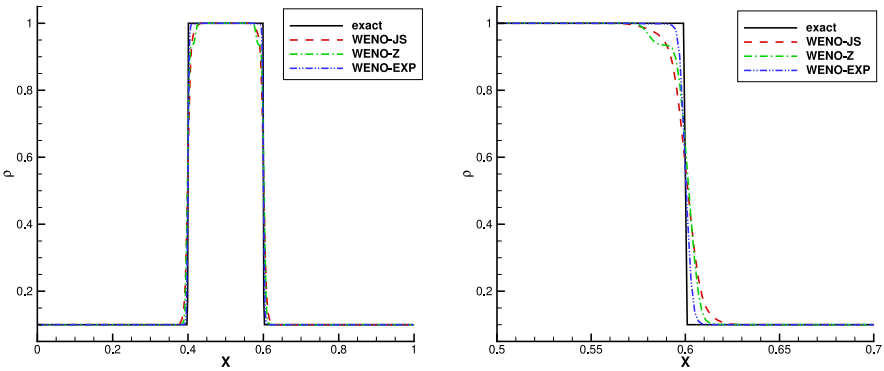
Table 2 Convergence order for the 1D smooth interface advection problem

cells	WENO-JS				WENO-Z				WENO-EXP			
	$L_\infty(z_1)$	order	$L_1(z_1)$	order	$L_\infty(z_1)$	order	$L_1(z_1)$	order	$L_\infty(z_1)$	order	$L_1(z_1)$	order
20	3.21e-3		2.12e-3		6.56e-4		3.69e-4		8.34e-5		4.65e-5	
40	1.33e-4	4.59	6.98e-5	4.93	1.92e-5	5.09	1.05e-5	5.14	1.11e-6	6.24	6.91e-7	6.07
80	4.52e-6	4.88	2.17e-6	5.01	5.82e-7	5.05	3.18e-7	5.04	1.69e-8	6.03	1.03e-8	6.06
160	1.43e-7	4.98	6.76e-8	5.01	1.79e-8	5.02	9.80e-9	5.02	2.58e-10	6.03	1.58e-10	6.03



(a) mixture density (b) mechanical equilibrium

Fig. 2 Numerical results for the one-dimensional smooth advection problem ($N = 40$)



(a) density ρ (b) zoomed-in view of ρ

Fig. 3 Numerical results for the one-dimensional sharp advection problem

fraction out of [0, 1]. While by using the BPPP limiter, admissible solutions are well preserved.

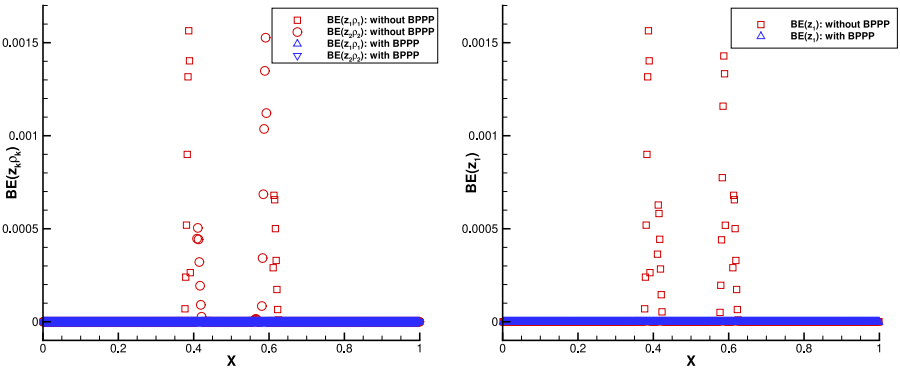
5.3 One-dimensional Gas-Water Shock Tube Problem

This case is a one-dimensional two-medium gas-water shock tube problem that contains a strong contact discontinuity. The initial conditions are set to be as follows

$$(\rho_1, \rho_2, u, p, z_1) = \begin{cases} (1.0, 200.0, 0.0, 1.0, 1.0 - 10^{-6}), & \text{if} \quad -1 < x \leq 0, \\ (1.0, 200.0, 0.0, 10000.0, 10^{-6}), & \text{if} \quad 0 < x < 1, \end{cases} \tag{56}$$

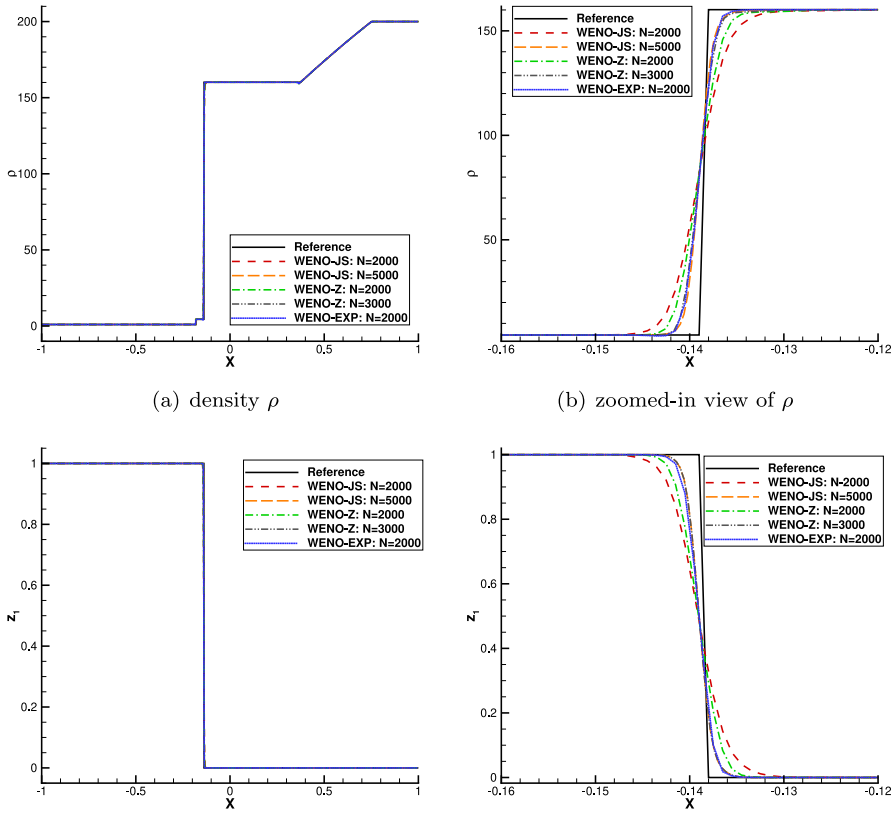
with $\gamma_1 = 1.4,\ \pi_{\infty,1} = 0.0$ and $\gamma_2 = 4.4,\ \pi_{\infty,2} = 6000.0$. The output time is $T = 0.04$.

We conduct the computation by using the WENO-JS, WENO-Z and WENO-EXP schemes. The numerical results shown in Fig. 5 demonstrate that the WENO-EXP scheme provides a obviously better solution than the WENO-JS and WENO-Z schemes under the same mesh resolution $N = 2000$, especially on the regions around the material interface.



(a) BE($z_k\rho_k$) (b) BE(z_1)

Fig. 4 Bound errors for the one-dimensional sharp advection problem



(a) density ρ (b) zoomed-in view of ρ

(c) volume fraction z_1 (d) zoomed-in view of z_1

Fig. 5 Numerical Results for the One-dimensional Gas-Water Shock Tube Problem

Table 3 CPU time for the one-dimensional gas-water shock tube problem

WENO-JS		WENO-Z		WENO-EXP
$N = 2000$	$N = 5000$	$N = 2000$	$N = 3000$	$N = 2000$
80.38s	501.96s	83.22s	184.20s	102.83s

Moreover, in order to achieve a comparable computational resolution, the WENO-JS and WENO-Z schemes require nearly 2.5 times ($N = 5000$) and 1.5 times ($N = 3000$) the number of mesh cells, respectively, which significantly improves the computational cost. Tab. 3 lists the CPU time taken by the three numerical schemes. We can see that although the WENO-EXP scheme costs more CPU time than the other two schemes under the mesh resolution, it demonstrates superior efficiency from the standpoint of equivalent computational resolution. Finally, we mention that due to the appearance of negative squared speed of sound, the computational will immediately blow up without using the bound-preserving and positivity-preserving limiter in this test case.

5.4 Two-dimensional Smooth Advection Problem

This case is a two-dimensional smooth advection problem with the initial condition given by

$$(\rho_1, \rho_2, u, v, p, z_1) = (1.0, 2.0, 1.0, 1.0, 1.0, 0.5 + 0.4999 \sin(2\pi(x + y))), \tag{57}$$

where $(x, y) \in [0, 1] \times [0, 1]$ with periodic boundary conditions, and the output time is $T = 1.0$. The material parameters are same as those used in one-dimensional case.

From Tab. 4 we observe that similar to the one-dimensional case, the present WENO-EXP scheme produces a clearly lower numerical error than the WENO-JS and WENO-Z schemes, and it is able to achieve the designed sixth-order of accuracy in solving the two-dimensional smooth advection problem. The numerical errors shown in Fig. 6 demonstrate that the mechanical equilibrium is also well maintained by the present scheme since the velocity and pressure errors are around $\mathcal{O}(10^{-12})$ while the velocity and pressure solutions are around $\mathcal{O}(1)$.

5.5 Richtmyer-Meshkov Instability

A simulation of the Richtmyer-Meshkov instability [38–41] is carried out in this case. The initial single-mode perturbation of an air-SF6 interface is generated by

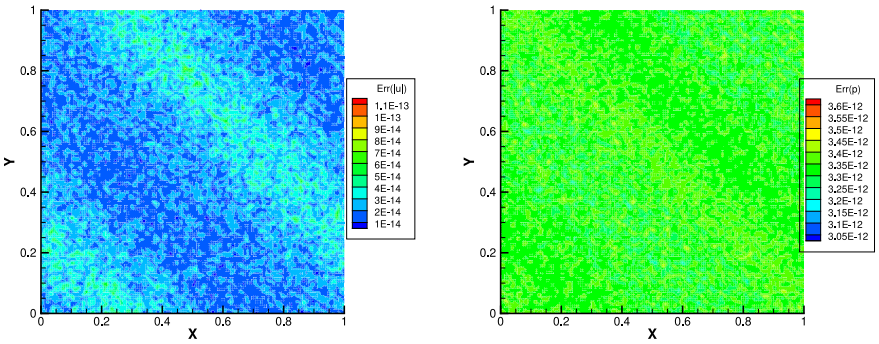
$$\Phi(x, y) = x - 2.9 + 0.1 \sin(2\pi(y - 0.25)), \quad (x, y) \in [0, 4] \times [0, 1]. \tag{58}$$

A planar shock wave with Mach number 1.24 is located at $x = 3.2$, and the initial conditions are

$$(\rho, u, v, p, \gamma, \pi_\infty) = \begin{cases} (5.04, 0.0, 0.0, 1.0, 1.093, 0.0), & \text{SF6}, \\ (1.0, 0.0, 0.0, 1.0, 1.4, 0.0), & \text{pre-shock air}, \\ (1.411, -0.39, 0.0, 1.628, 1.4, 0.0), & \text{post-shock air}. \end{cases} \tag{59}$$

Table 4 Convergence order for the 2D smooth interface advection problem.

cells	WENO-JS				WENO-Z				WENO-EXP			
	$L_\infty(z_1)$	order	$L_1(z_1)$	order	$L_\infty(z_1)$	order	$L_1(z_1)$	order	$L_\infty(z_1)$	order	$L_1(z_1)$	order
20×20	6.00e-3		4.26e-3		1.71e-3		9.85e-4		5.06e-4		3.41e-4	
40×40	2.47e-4	4.60	1.43e-4	4.89	4.49e-5	5.25	2.48e-5	5.31	8.59e-6	5.88	5.25e-6	6.02
80×80	8.71e-6	4.83	4.40e-6	5.03	1.26e-6	5.15	6.96e-7	5.15	1.29e-7	6.06	8.16e-8	6.01
160×160	2.81e-7	4.95	1.36e-7	5.01	3.73e-8	5.08	2.05e-8	5.08	2.13e-9	5.92	1.27e-9	6.01



(a) velocity error (b) pressure error

Fig. 6 Numerical errors for the two-dimensional smooth advection problem ($N = 160 \times 160$)

Reflection boundary conditions are used at the bottom and upper boundaries, while extrapolation at the left boundary and post-shock values at the right boundary, that is

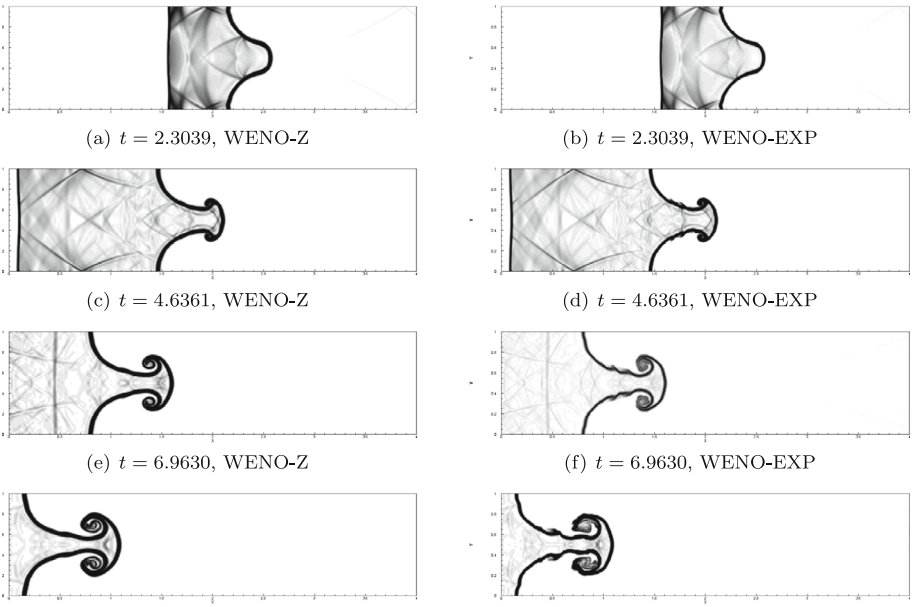
$$\begin{cases} (\rho, u, v, p)|_{y_b^-} = (\rho, u, -v, p)|_{y_b^+}, & \text{at the bottom boundary } y_b = 0, \\ (\rho, u, v, p)|_{y_b^+} = (\rho, u, -v, p)|_{y_b^-}, & \text{at the upper boundary } y_b = 1, \\ (\rho, u, v, p)|_{x_b^-} = (\rho, u, v, p)|_{x_b^+}, & \text{at the left boundary } x_b = 0, \\ (\rho, u, v, p)|_{x_b^+} = (1.411, -0.39, 0.0, 1.628), & \text{at the right boundary } x_b = 4. \end{cases} \tag{60}$$

The computation is conducted by using the WENO-Z and WENO-EXP schemes with $N = 800 \times 200$ uniform mesh cells. Fig. 7 shows the flow evolution by plotting the density gradient generated with $\phi = \exp(|\nabla\rho|/|\nabla\rho|_{\max})$. The formation of rolled-up vortical structure and further distortions of the interface after the incident shock wave hitting the interface can be observed from the figure. The flow features are quite similar to those reported in [40]. Moreover, the present WENO-EXP scheme produces a clearly sharper material interface than the WENO-Z scheme, and it also produces more small-scale distortions of the interface.

Fig. 8 presents the time evolution of the location of the spike, the leading edge of the bubble, and the thickness of the mixing layer. Here, the thickness of the mixing layer is defined as a distance between the spike and the bubble as illustrated by Fig. 8(a). It can be seen that the numerical results computed by the two schemes agree well with the reference result [40], especially the evolution of the thickness of the mixing layer is quite similar.

5.6 Interaction of a Shock Wave with a Helium Bubble in Air

This case considers the interaction between a shock wave with Mach number 1.22 and a helium bubble in air, which has been widely investigated in literatures to demonstrate the performance of the numerical schemes [9, 13, 40, 41]. At the initial time, the diameter of the helium bubble is 1.0 and its center locates at $(3.5, 0.0)$ in the computational domain $[0, 6.5] \times [-0.89, 0.89]$. The left-moving shock wave is located at $x = 4.5$, and the initial values of the fluid variables are



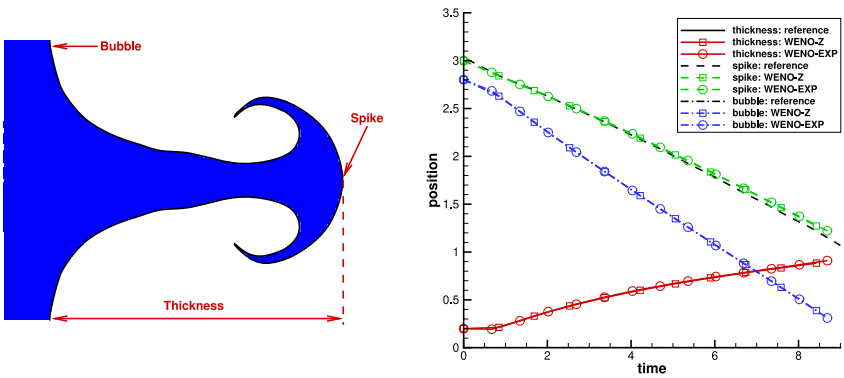
(a) $t = 2.3039$, WENO-Z (b) $t = 2.3039$, WENO-EXP

(c) $t = 4.6361$, WENO-Z (d) $t = 4.6361$, WENO-EXP

(e) $t = 6.9630$, WENO-Z (f) $t = 6.9630$, WENO-EXP

(g) $t = 9.2433$, WENO-Z (h) $t = 9.2433$, WENO-EXP

Fig. 7 Density gradients for the Richtmyer-Meshkov instability



(a) schematic of spike/bubble/thickness (b) position of spike/bubble/thickness

Fig. 8 Characteristic points position for Richtmyer-Meshkov instability

$$(\rho, u, v, p, \gamma, \pi_\infty) = \begin{cases} (0.1819, 0.0, 0.0, \dfrac{1.0}{1.4}, 1.648, 0.0), & \text{helium bubble,} \\ (1.0, 0.0, 0.0, \dfrac{1.0}{1.4}, 1.4, 0.0), & \text{pre-shock air,} \\ (1.3764, -0.3336, 0.0, \dfrac{1.5698}{1.4}, 1.4, 0.0), & \text{post-shock air.} \end{cases} \tag{61}$$

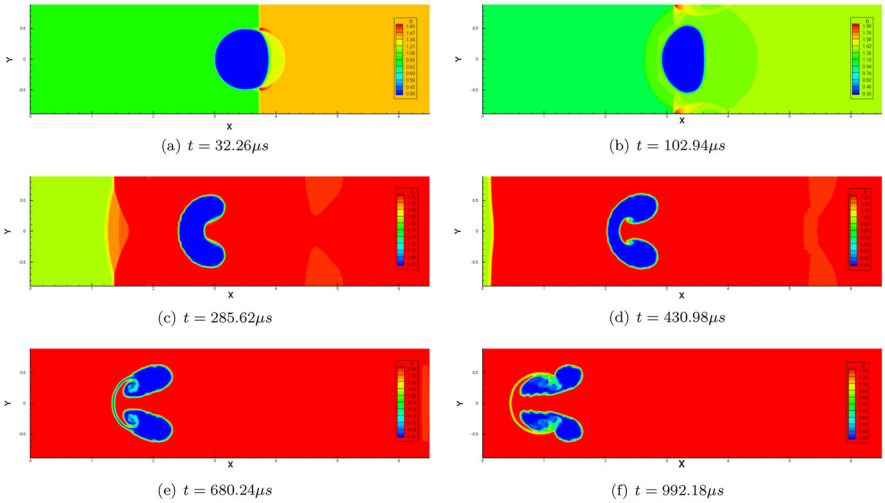


Fig. 9 Density profile for the shock wave-helium bubble interaction in air

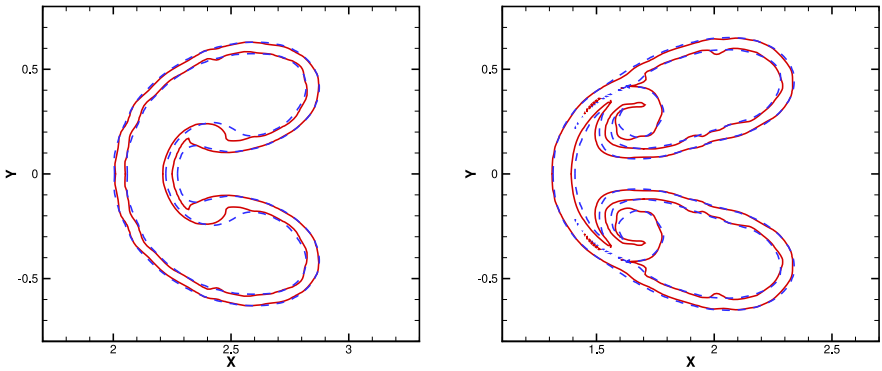
Reflection boundary conditions are used at the bottom and upper boundaries of the domain, while extrapolation at the left boundary and post-shock values at the right boundary. The computational domain is divided into a total of $N = 650 \times 178$ uniform mesh cells.

Fig. 9 shows the density profile at several physical time instants computed by the WENO-EXP scheme. It can be seen that the complex small wave structures are clearly captured by the present scheme and the major features are in well agreement with those reported in references [40, 41]. The comparison of the material interfaces resolved by the WENO-Z and WENO-EXP schemes is shown in Fig. 10. Here, the material interface corresponds to the zone where the volume fraction changes rapidly from 0.05 to 0.95. Obviously, the evolving structure of the material interface resolved by the present scheme is more intricate and detailed than that of the WENO-Z scheme. It clearly illustrates that the present scheme possesses the capabilities of higher resolution and better scale-resolving. In Fig. 11, we plot a quantitative comparison of three characteristic points of the helium bubble, including the Jet, Downstream and Upstream points. As shown in the figure, the numerical results agree well with the reference one reported in [41].

Next, we test the performance of the BPPP limiter. Fig. 12 shows the distribution of non-physical cells, in which the BPPP limiter is active. We observe that the nonphysical cells mainly locate around the material interface, which may trigger a failure of the computation. However, with the use of the BPPP limiter, positive partial density, positive internal energy and bounded volume fraction are well preserved. Therefore, the present bound-preserving and positivity-preserving limiting algorithm plays an essential role in improving the computational robustness.

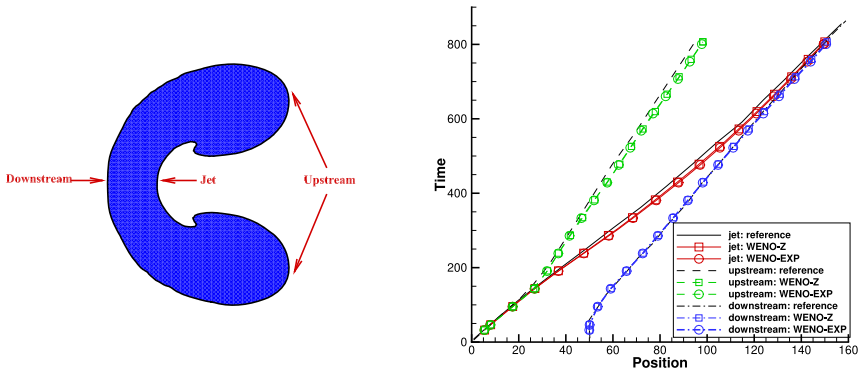
5.7 Interaction of a Shock Wave with a Gas Bubble in Water

We consider the interaction between a shock wave with Mach number 1.72 and a gas bubble in water. Similar test case has been previously investigated in [4, 10, 11, 40, 42, 43], and an experimental test was also carried out in [44]. At the initial time, the diameter of the gas



(a) $t = 430.98\mu s$ (b) $t = 680.24\mu s$

Fig. 10 Material interface for the shock wave-helium bubble interaction in air (dashed line: WENO-Z; solid line: WENO-EXP)



(a) schematic of downstream/jet/upstream (b) position of downstream/jet/upstream

Fig. 11 Characteristic points position for the shock wave-helium bubble interaction in air

bubble is 6 and its center locates at (6, 0) in the computational domain $[0, 12] \times [-6, 6]$. The left-moving shock wave is located at $x = 11.4$. The initial values of the fluid variables are given by

$$(\rho, u, v, p, \gamma, \pi_\infty) = \begin{cases} (0.0012, 0.0, 0.0, 1.0, 1.4, 0.0), & \text{gas bubble,} \\ (1.0, 0.0, 0.0, 1.0, 4.4, 6000.0), & \text{pre-shock water,} \\ (1.325, -68.525, 0.0, 19153.0, 4.4, 6000.0), & \text{post-shock water.} \end{cases} \tag{62}$$

Reflection boundary conditions are used at the bottom and upper boundaries of the domain, while extrapolation at the left boundary and post-shock values at the right boundary. The computational domain is divided into a total of $N = 800 \times 800$ uniform mesh cells.

We show the numerical results obtained by the WENO-EXP scheme by plotting the density profiles at six time instants in Fig. 13. We see that after hitting the gas bubble, the incident shock wave transmits a relatively weak shock into the gas bubble and a strong reflected rarefaction wave into the water. Along with time, a water jet impacts the gas bubble, and the gas bubble becomes involuted and is cut into two parts at approximately $t = 0.0368$. At the

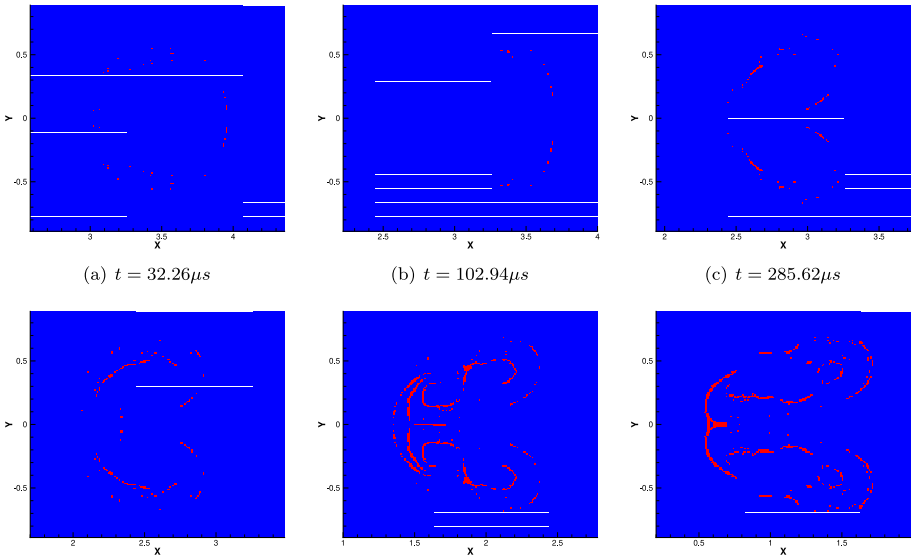


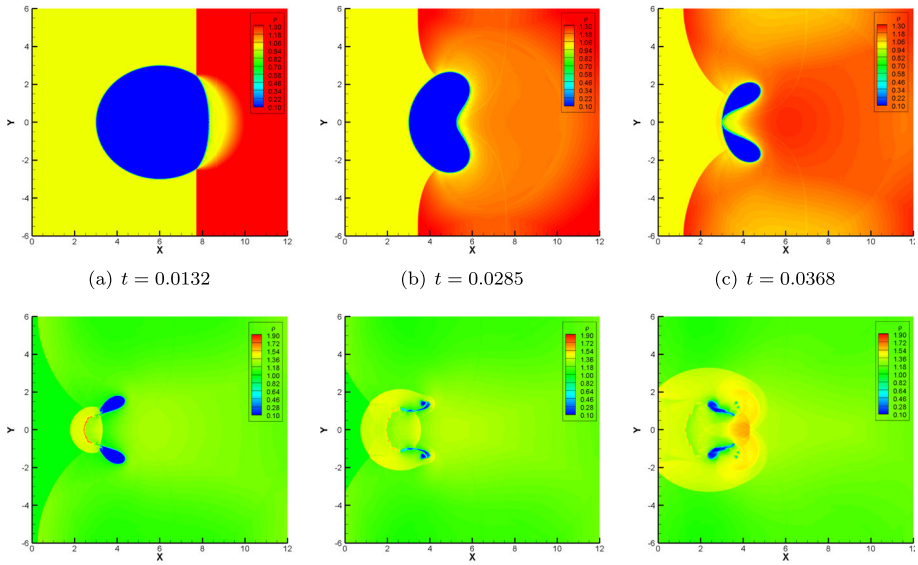
Fig. 12 Nonphysical cells distribution for the shock wave-helium bubble interaction in air

meanwhile, an intense blast wave is generated which eventually causes the collapse of the gas bubble. These main features are well captured by the present scheme and also agree well with previous results reported in [40, 42]. In order to demonstrate the high-resolution property of the present scheme, we conduct the computation again by using the WENO-Z scheme and show the comparison of the material interfaces resolved by the two schemes in Fig. 14. Compared to the WENO-Z scheme, a obviously sharper material interface is generated by the WENO-EXP scheme. Specifically, the thickness of the material interface along the normal direction reduces nearly by half by using the present WENO-EXP scheme, which validates its superiority in improving the computational resolution in the regions involving large gradients.

Fig. 15 shows the time evolution of the volume fraction contour $z_1 = 0.5$ computed by the WENO-EXP scheme, from which we can clearly observe the deformation process of the gas bubble induced by the passage of the shock wave. A quantitative result of the bubble size, including the bubble height and width, computed by the WENO-Z and WENO-EXP schemes is also presented in this figure, which shows that the numerical results are in well consistent with the reference one [42]. Finally, we present the distribution of nonphysical cells obtained by the WENO-EXP scheme in Fig. 16. Similar to the previous test case, the nonphysical cells mainly locate in the vicinity of the material interface where involves large ratios of density and pressure. Without using the bound-preserving and positivity-preserving limiter, the nonphysical numerical solutions will severely cause a failure of the computation at the very early stage of the computation.

5.8 Underwater Explosion

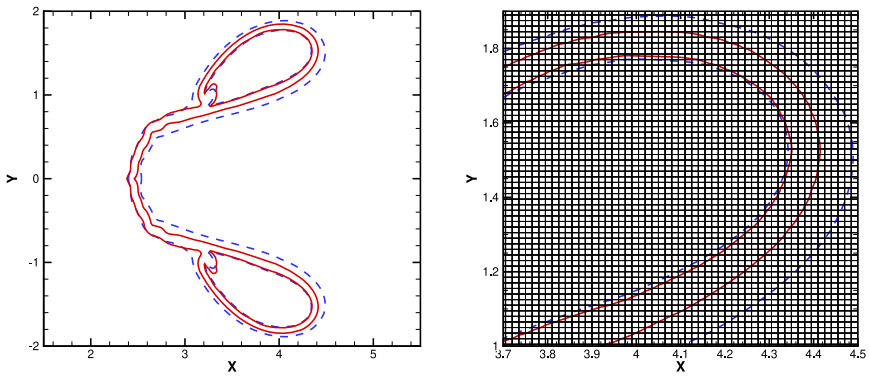
The last case is the model of underwater explosion [10, 43, 45, 46]. The solution domain is $[-2, 2] \times [-1.5, 1.5]$, and the air and water are separated by a horizontal water surface locates at $y = 0$. Below the water surface, there is a highly pressurized explosion gas cylinder with



(a) $t = 0.0132$ (b) $t = 0.0285$ (c) $t = 0.0368$

(d) $t = 0.0405$ (e) $t = 0.0440$ (f) $t = 0.0486$

Fig. 13 Density profile for the shock wave-gas bubble interaction in water



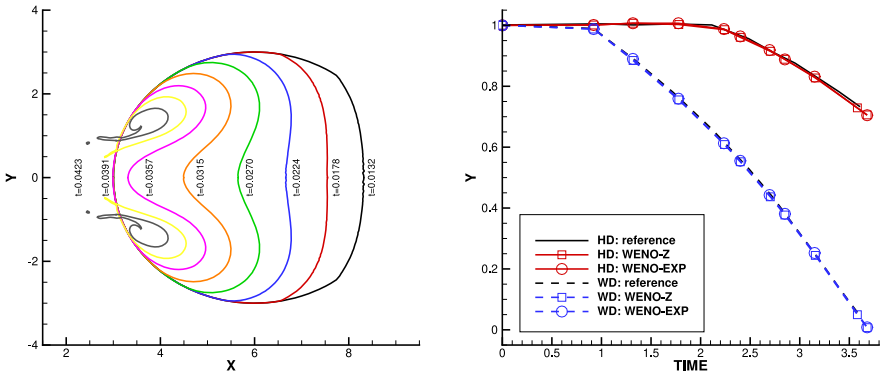
(a) $t = 0.0405$ (b) zoomed-in view

Fig. 14 Material interface for the shock wave-gas bubble interaction in water (dashed line: WENO-Z; solid line: WENO-EXP)

the center locates at $(0, -0.3)$ and a radius $r = 0.12$. Initial condition of the state variables are given by

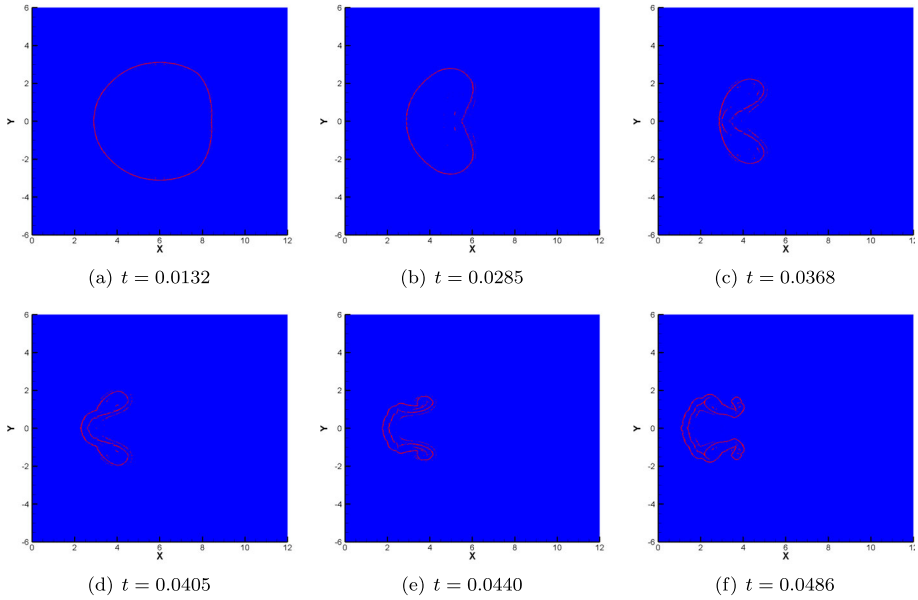
$$(\rho, u, v, p, \gamma, \pi_\infty) = \begin{cases} (1.25, 0.0, 0.0, 10000.0, 1.4, 0.0), & \text{gas cylinder}, \\ (0.001225, 0.0, 0.0, 1.01325, 1.4, 0.0), & \text{air}, \\ (1.0, 0.0, 0.0, 1.01325, 4.4, 6000.0), & \text{water}. \end{cases} \tag{63}$$

Reflecting boundary condition is imposed for the upper wall, and extrapolation boundary condition is imposed for the other three walls.



(a) volume fraction $z_1 = 0.5$, WENO-EXP (b) height (HD) and width (WD)

Fig. 15 Results for the shock wave-gas bubble interaction in water



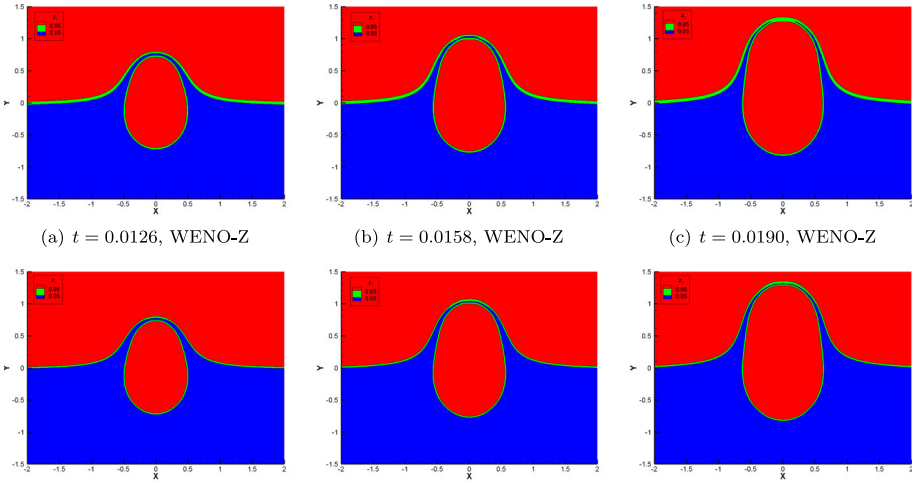
(a) $t = 0.0132$ (b) $t = 0.0285$ (c) $t = 0.0368$

(d) $t = 0.0405$ (e) $t = 0.0440$ (f) $t = 0.0486$

Fig. 16 Nonphysical cells distribution for the shock wave-gas bubble interaction in water

Fig. 17 shows the time evolutions of the material interface computed by the WENO-Z and WENO-EXP schemes with $N = 800 \times 600$ uniform mesh cells. We can see that compared to the WENO-Z scheme, a much sharper material interface is resolved by the WENO-EXP scheme. Besides, the material interface around the top point of the gas cylinder is severely smeared out by the WENO-Z scheme at the late stage of the computation, while the WENO-EXP scheme can still provide a thin water bridge between the expanding gas cylinder and the ambient air. The density profiles along $x = 0$ computed by the two numerical schemes are shown in Fig. 18, which exhibit good agreement with the reference one [45].

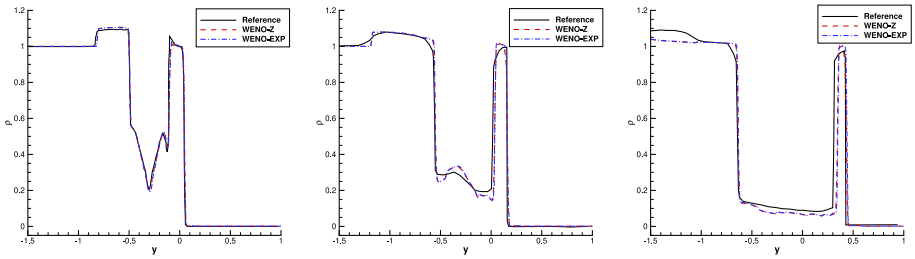
Since this test case involves a large density ratio about $1000 : 1$ and a large pressure ratio about $10000 : 1$, nonphysical numerical solutions will appear during the computation. Fig. 19



(a) $t = 0.0126$, WENO-Z (b) $t = 0.0158$, WENO-Z (c) $t = 0.0190$, WENO-Z

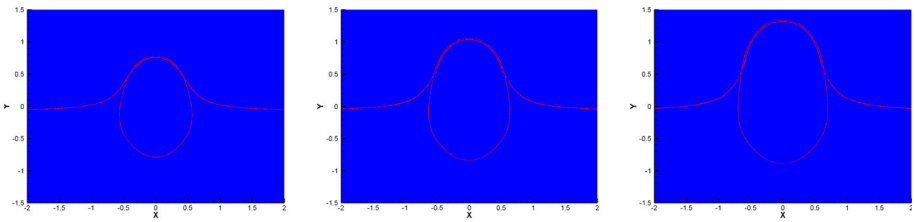
(d) $t = 0.0126$, WENO-EXP (e) $t = 0.0158$, WENO-EXP (f) $t = 0.0190$, WENO-EXP

Fig. 17 Material interface evolution for the underwater explosion



(a) $t = 0.0020$ (b) $t = 0.0040$ (c) $t = 0.0080$

Fig. 18 Density profile along $x = 0$ for the underwater explosion



(a) $t = 0.0126$ (b) $t = 0.0158$ (c) $t = 0.0190$

Fig. 19 Nonphysical cells distribution for the underwater explosion

shows the distribution of nonphysical cells obtained by the WENO-EXP scheme at several time instants. The location of nonphysical cells are consistent with the shape of material interface, and the computation will immediately blow up without using the bound-preserving and positivity-preserving limiter. This clearly demonstrates that the present bound-preserving and positivity-preserving limiter is effective in maintaining the intrinsic boundedness for the numerical solutions, which thereby significantly improves the computational robustness.

6 Conclusions

This work presented a finite volume WENO scheme based on the exponential approximation space for solving the five-equation transport model of compressible two-medium flows. In order to improve the computational performance, we conducted a detailed study on deriving the shape parameter formulation which can enhance one order of accuracy without increasing the stencil size and meanwhile maintaining non-negative linear weights of the WENO reconstruction. More importantly, we proposed a specific strategy for designing the bound-preserving and positivity-preserving limiting algorithm targeted for non-polynomial approximation space based numerical schemes. The present scheme is also able to maintain the mechanical equilibrium which is of vital importance in maintaining an oscillation-free material interface. Future work will focus on its extension to compressible multi-medium flows with more general equation of state.

Appendix A Cell-wise Characteristic Projection Based WENO Reconstruction

In this appendix, we will prove that the relationship between the two-dimensional cell average and the reconstructed solutions at the quadrature points (49) still holds true through using the cell-wise characteristic projection. The proving process at one dimension is similar.

Consider the target cell $I_{i,j} \in \Omega_h$, we denote the left and right eigenmatrix based on $\overline{\mathbf{Q}}_{i,j}$ along x/y-direction as $\mathbf{L}_{x/y}(\overline{\mathbf{Q}}_{i,j})$ and $\mathbf{R}_{x/y}(\overline{\mathbf{Q}}_{i,j})$, respectively. The main procedure of the cell-wise characteristic projection based WENO reconstruction contains the following steps:

1. Project the cell average into the characteristic space along y-direction by using

$$\overline{\mathbf{Q}}^{(c)}_{i+l,j+r} = \mathbf{L}_y(\overline{\mathbf{Q}}_{i+l,j})\overline{\mathbf{Q}}_{i+l,j+r}, \quad l, r = -2, \cdots, 2. \tag{64}$$

2. Perform the WENO reconstruction procedure to obtain $\mathbf{Q}^{(c)}_{i+l}(y) \in \mathbb{E}_5(y)$ that satisfies

$$\frac{1}{\Delta y}\int_{y_{j+r-\frac{1}{2}}}^{y_{j+r+\frac{1}{2}}} \mathbf{Q}^{(c)}_{i+l}(y)\mathrm{d}y = \overline{\mathbf{Q}}^{(c)}_{i+l,j+r}. \tag{65}$$

   Note that the integrals of the reconstructed solutions are approximated by the four-point Gauss-Lobatto quadrature rule (44) which implies the following equality is satisfied

$$\overline{\mathbf{Q}}^{(c)}_{i+l,j} \equiv \sum_{\beta=1}^{4} \frac{\widehat{\omega}_\beta}{2}\mathbf{Q}^{(c)}_{i+l}(\widehat{y}^\beta_j). \tag{66}$$

3. Project the reconstructed solutions at the quadrature points in y back into the physical space

$$\mathbf{Q}_{i+l}(\widehat{y}^\beta_j) = \mathbf{R}_y(\overline{\mathbf{Q}}_{i+l,j})\mathbf{Q}^{(c)}_{i+l}(\widehat{y}^\beta_j). \tag{67}$$

   Combining (64), (66) and (67), it leads to that

$$\overline{\mathbf{Q}}_{i+l,j} \equiv \sum_{\beta=1}^{4} \frac{\widehat{\omega}_\beta}{2}\mathbf{Q}_{i+l}(\widehat{y}^\beta_j). \tag{68}$$

4. Project the reconstructed solutions onto the characteristic space along x-direction by using
$$\widetilde{\mathbf{Q}}_{i+l}^{(c)}(\widehat{y}_j^\beta) = \mathbf{L}_x(\overline{\mathbf{Q}}_{i,j})\mathbf{Q}_{i+l}(\widehat{y}_j^\beta). \tag{69}$$
5. Perform the WENO reconstruction procedure to obtain $\widetilde{\mathbf{Q}}_{i,j}^{(c)}(x, \widehat{y}_j^\beta) \in \mathbb{E}_5(x)$ that satisfies
$$\frac{1}{\Delta x}\int_{x_{i+l-\frac{1}{2}}}^{x_{i+l+\frac{1}{2}}} \widetilde{\mathbf{Q}}_{i,j}^{(c)}(x, \widehat{y}_j^\beta)\mathrm{d}x = \widetilde{\mathbf{Q}}_{i+l}^{(c)}(\widehat{y}_j^\beta). \tag{70}$$
Similarly, the integrals of the reconstructed solutions are calculated by (44) which gives
$$\widetilde{\mathbf{Q}}_{i}^{(c)}(\widehat{y}_j^\beta) \equiv \sum_{\alpha=1}^{4} \frac{\widehat{\omega}_\alpha}{2}\widetilde{\mathbf{Q}}_{i,j}^{(c)}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta). \tag{71}$$
6. Project the reconstructed solutions at the quadrature points back into the physical space
$$\mathbf{Q}_{i,j}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta) = \mathbf{R}_x(\overline{\mathbf{Q}}_{i,j})\widetilde{\mathbf{Q}}_{i,j}^{(c)}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta). \tag{72}$$
Combining (68), (69), (71) and (72), we have
$$\overline{\mathbf{Q}}_{i,j} \equiv \sum_{\alpha=1}^{4}\sum_{\beta=1}^{4} \frac{\widehat{\omega}_\alpha\widehat{\omega}_\beta}{4}\mathbf{Q}_{i,j}(\widehat{x}_i^\alpha, \widehat{y}_j^\beta). \tag{73}$$

This completes the cell-wise characteristic projection based WENO reconstruction, from which we also verify that by using the cell-wise characteristic projection, the cell average can still be decomposed into a convex combination of the reconstructed solutions at the quadrature points provided that the integrals of the reconstructed solutions are calculated by the quadrature rule.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

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